

Design of thermal heat storage systems adapted to hydrogen production by means of computational fluid dynamics

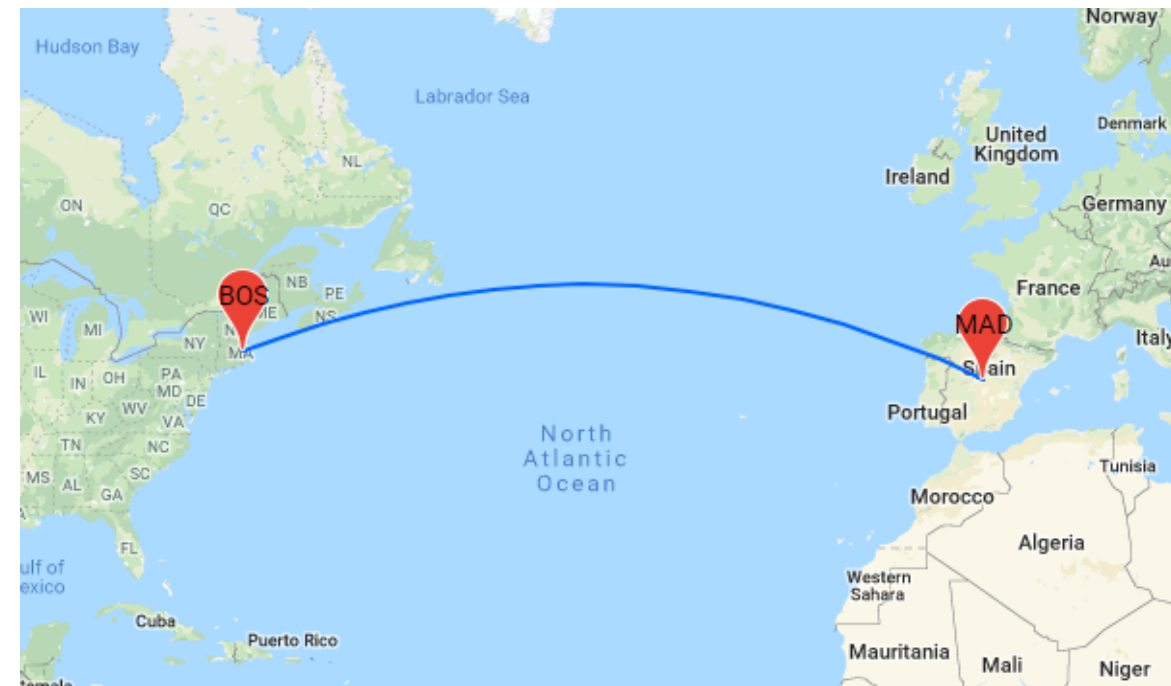
IMDEA Energy: High Temperature Processes Unit

Ottavia Personeni

Before we begin...

Ottavia:

- Mechanical Engineering & Computer Science (Learning Machines and Physical Systems)
- MISTI (MIT International Science and Technology Initiatives) Program: Boston -> Madrid
- IMDEA Project



[[Fly4Free](#)]

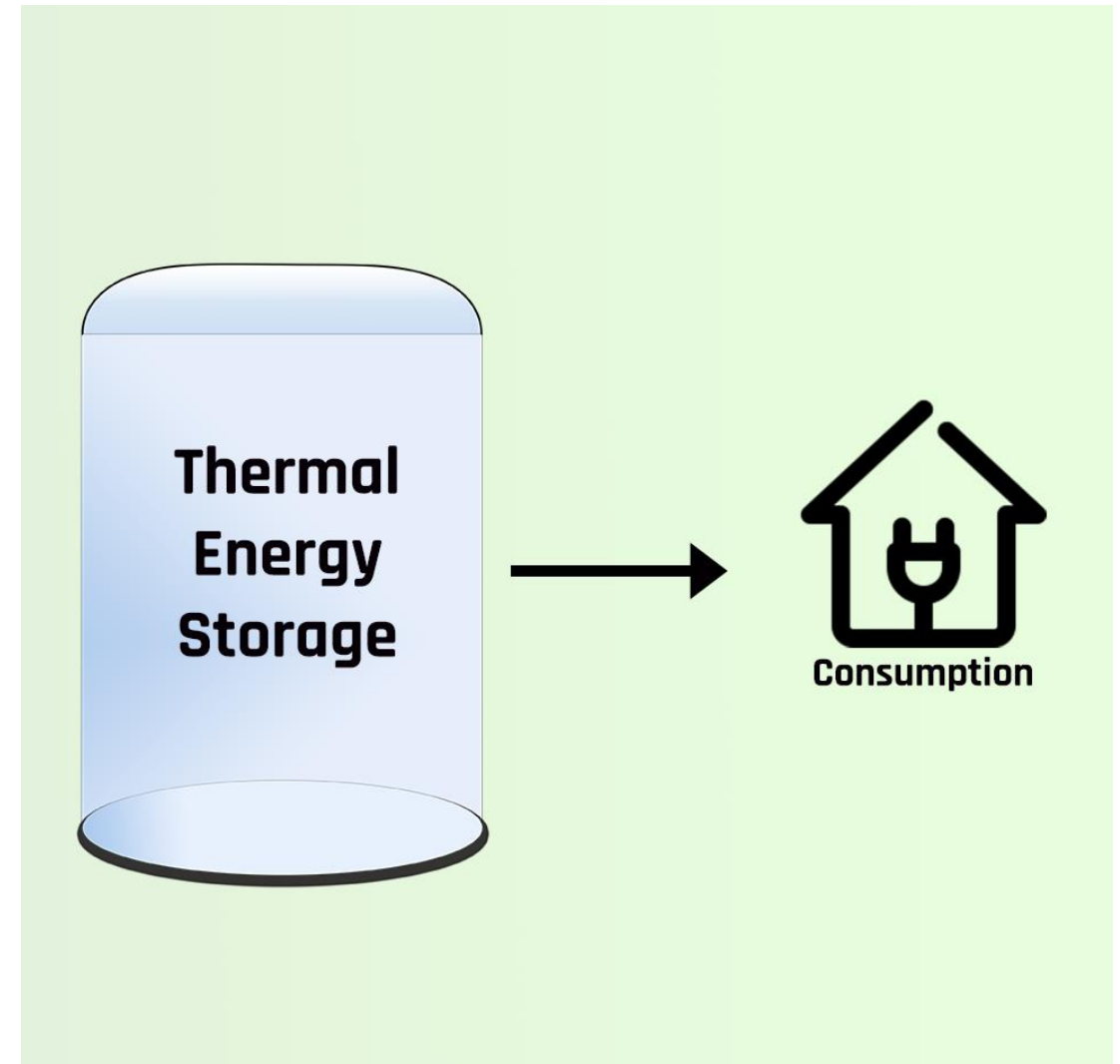
Overview

- I. Thermal Energy Storage
- II. Latent Thermal Energy Storage and Phase Change Materials
- III. CFD Modeling
- IV. PCM Simulations
- V. CFD Model Validation Experiment
- VI. Madrid Personal Experience
- VII. Final Remarks

I. Thermal Energy Storage (TES)

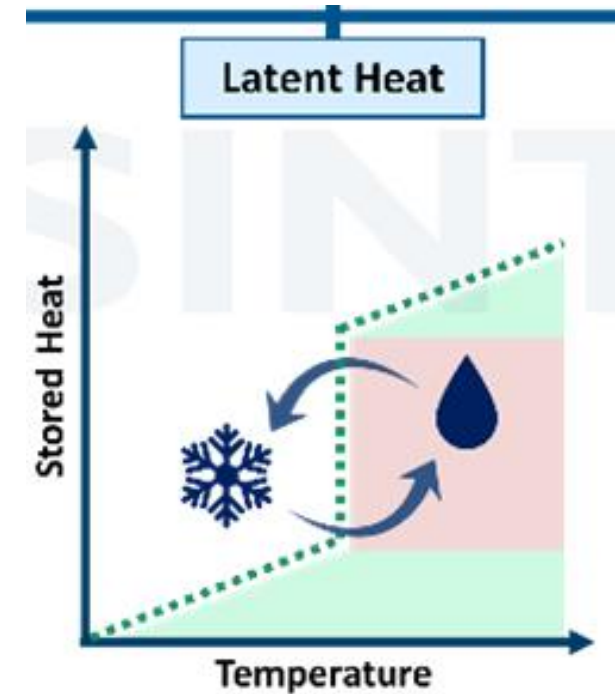
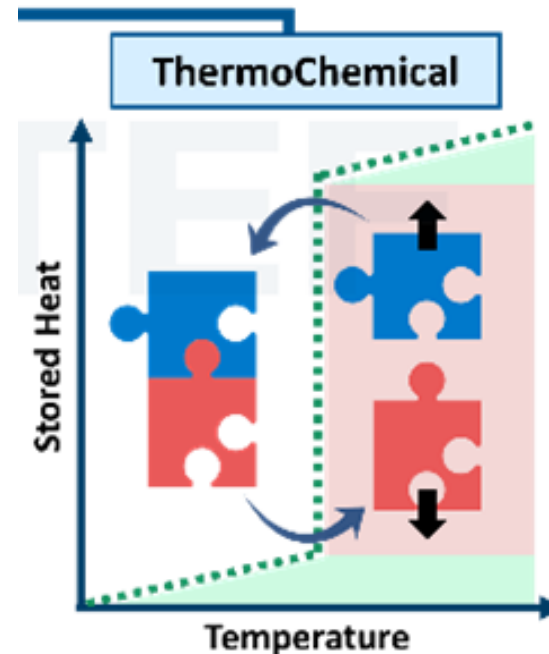
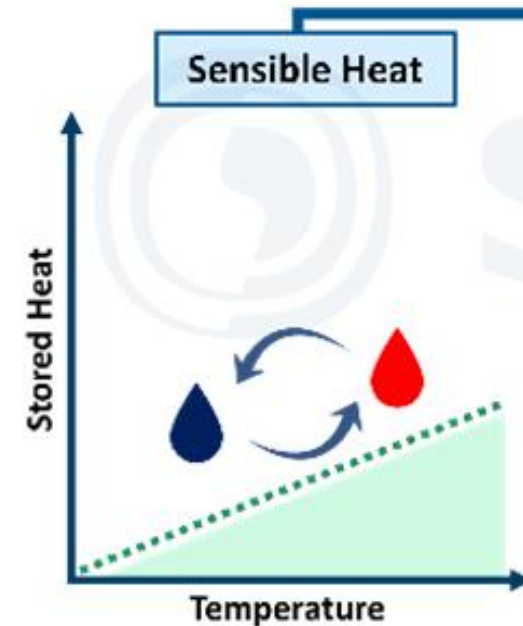
Why TES?

- Climate change and supply & demand concerns
- Solar thermal systems can address this issue
- **TES:** energy is stored until it is required



TES Types

- **Sensible** TES: transfer of heat to a body or material
- **Latent** TES: phase change at constant temperature
- **Thermochemical** TES: stored as heat of a chemical reaction

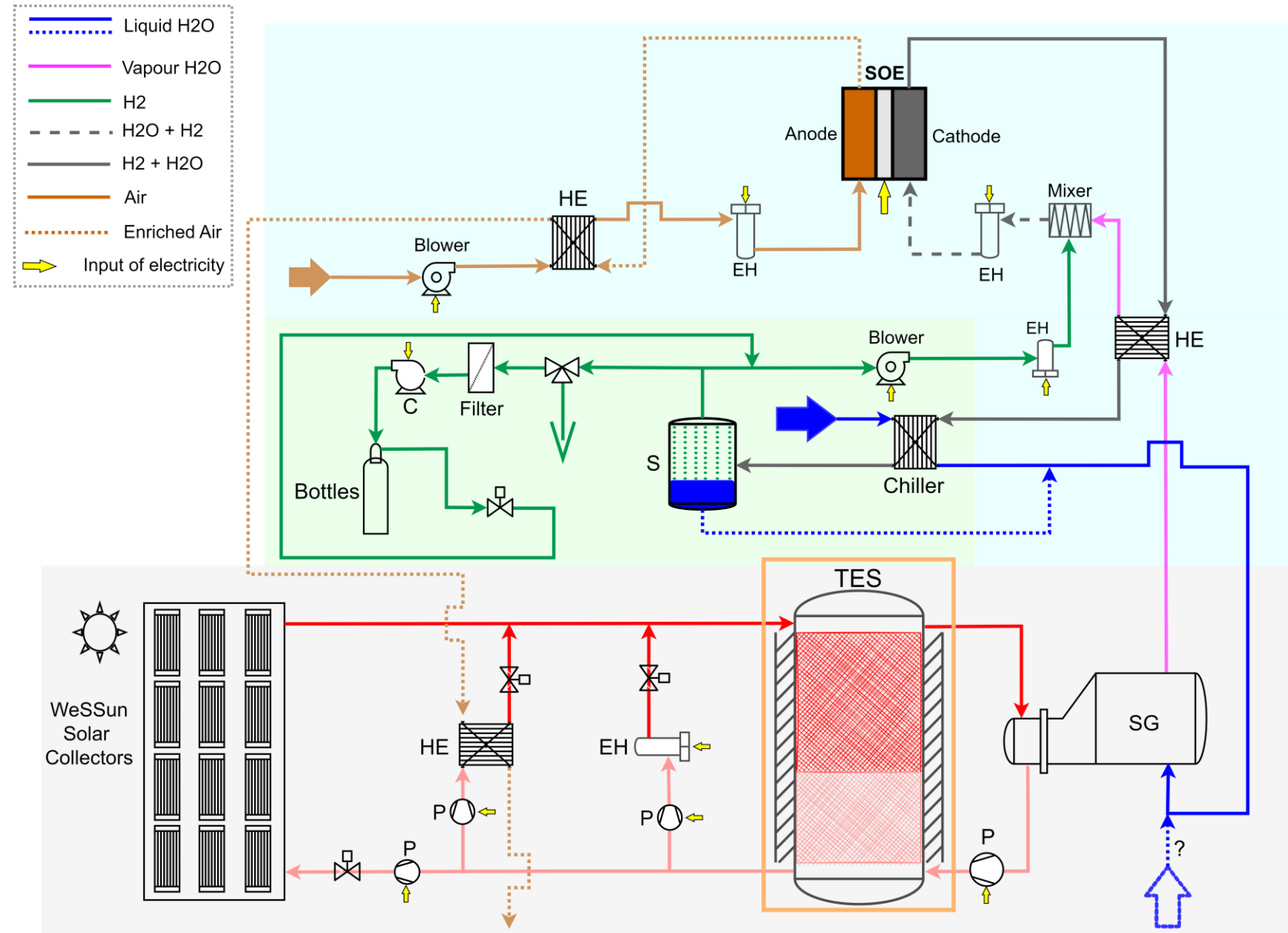


[[Beceiro, 2022](#)]

Green Hydrogen

- TES
- Latent TES

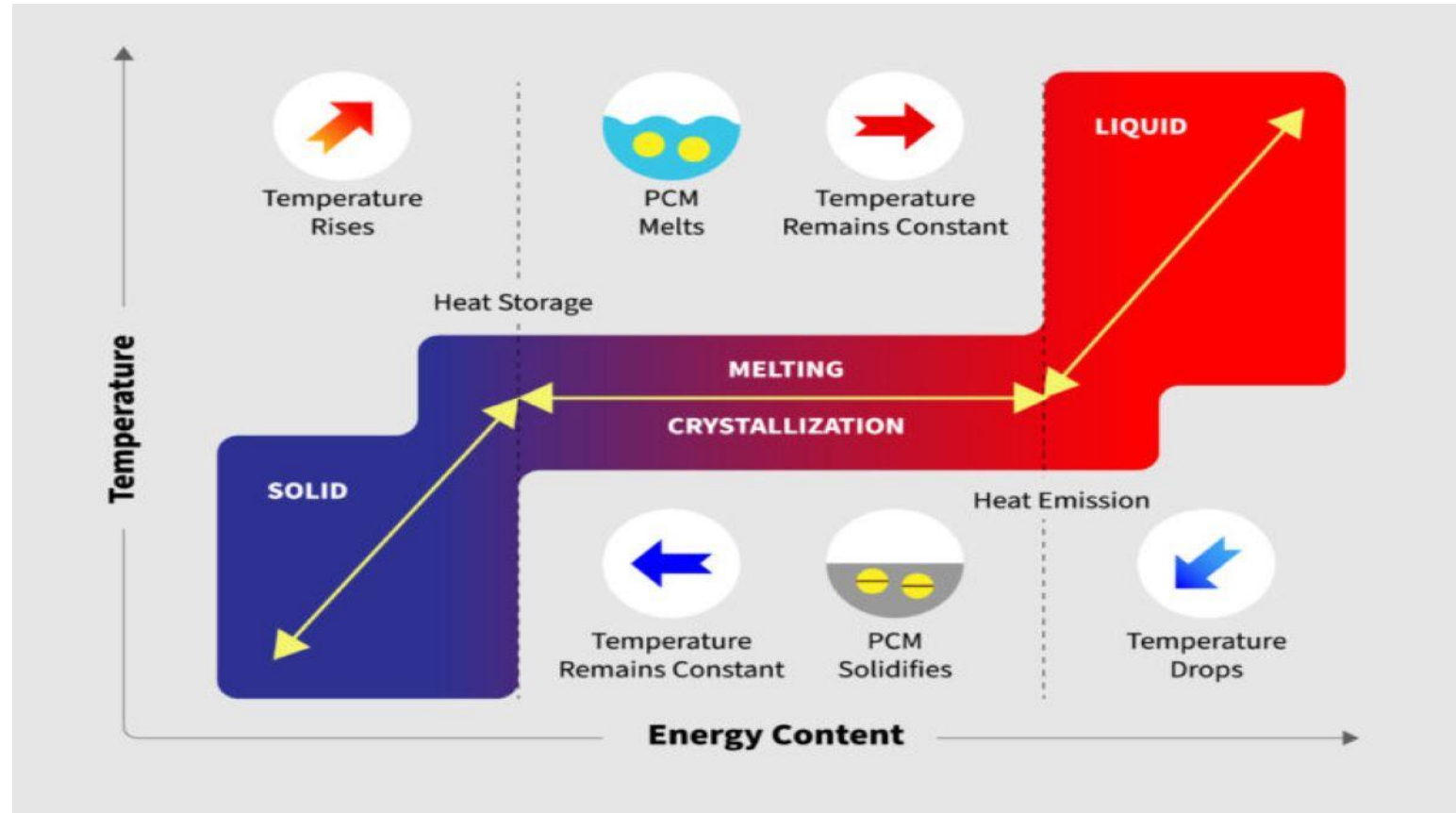
I. Thermal Energy Storage (TES)



Latent Thermal Energy Storage (LTES)

LTES and its Parts

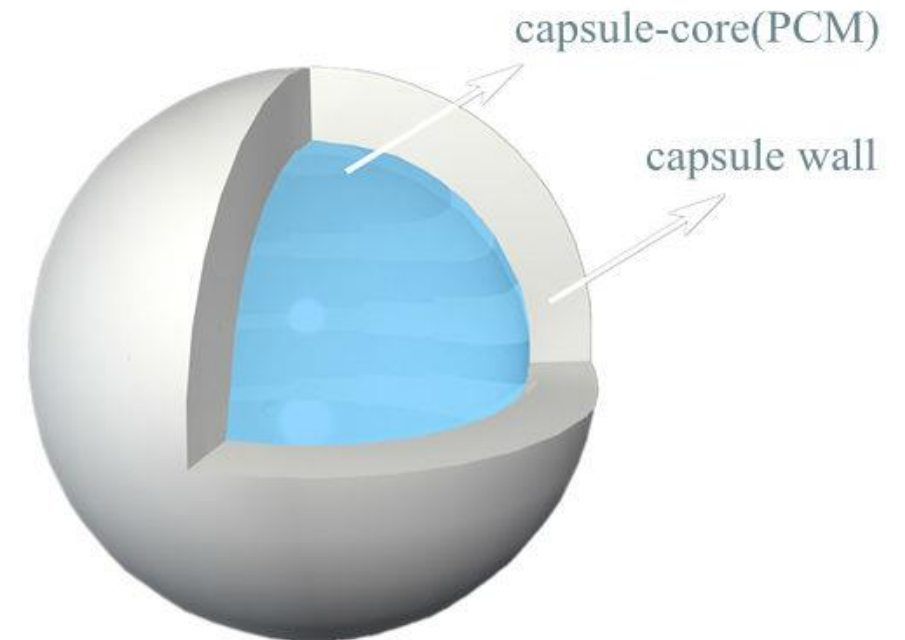
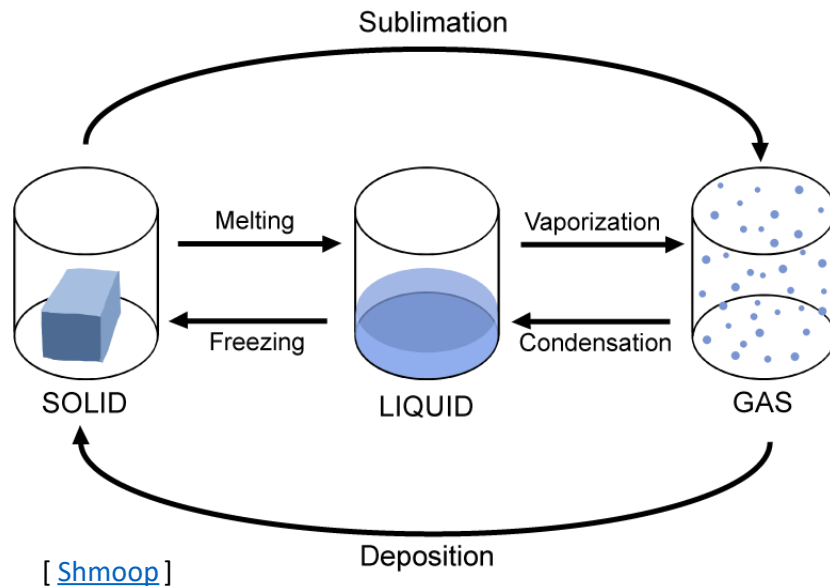
- Latent TES is **efficient**.
- Main limitation: the Phase Change Materials (**PCM**) need to meet many parameters



[[Mishra, 2015](#)]

PCM Capsules

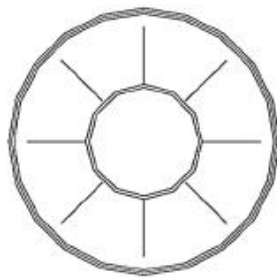
- Properties and types of PCMs include:
 - Organic PCMs
 - Salt Hydrates
 - Eutectic PCMs (lower melting point)



[[Kosny](#)]

Enhancements

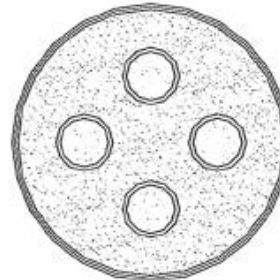
- **Low thermal conductivity** can be addressed with various **heat transfer enhancement** methods.



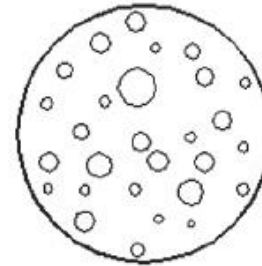
(i) Longitudinal or axial fins



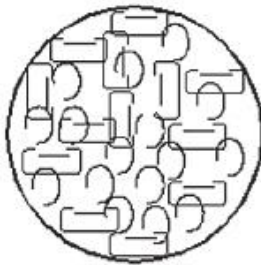
(ii) Circular fins



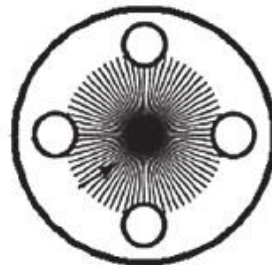
(iii) Multitubes or shell and tube



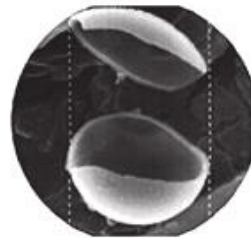
(iv) Bubble agitation



(v) Metal Rings



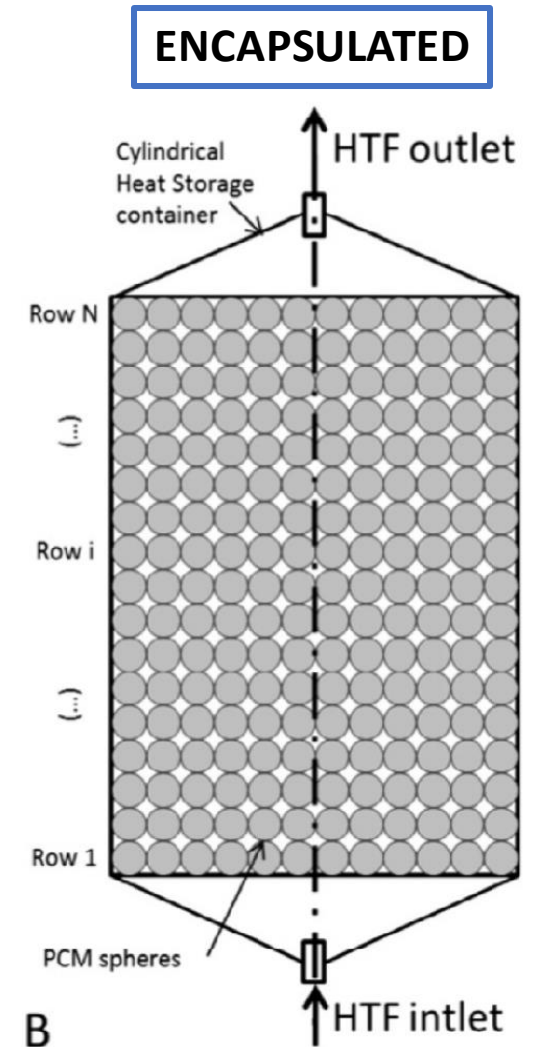
(vi) Multitubes and carbon brushes



(vii) Encapsulation



(ix) Metal Matrix



Spherical, Cylindrical, & Micro Capsules

- **Spherical capsules:** closer packing, better heat transfer.
- **Cylindrical capsules:** poorer convective heat transfer.
- **Micro capsules:** increase the natural convection heat transfer.



Our Choices

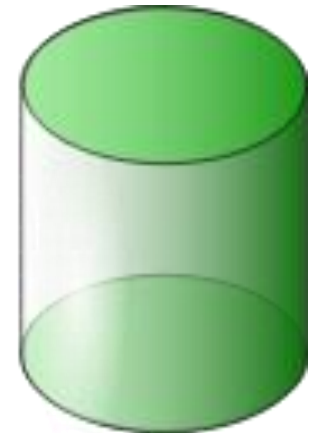
- **Latent** TES: more energy, constant temperature.
- **Encapsulated** systems: are more feasible/versatile.
- Proposed **PCM**: investigating Urea and Molten Salt.
- Effect of geometry.



[[Le Guillou, 2010](#)]

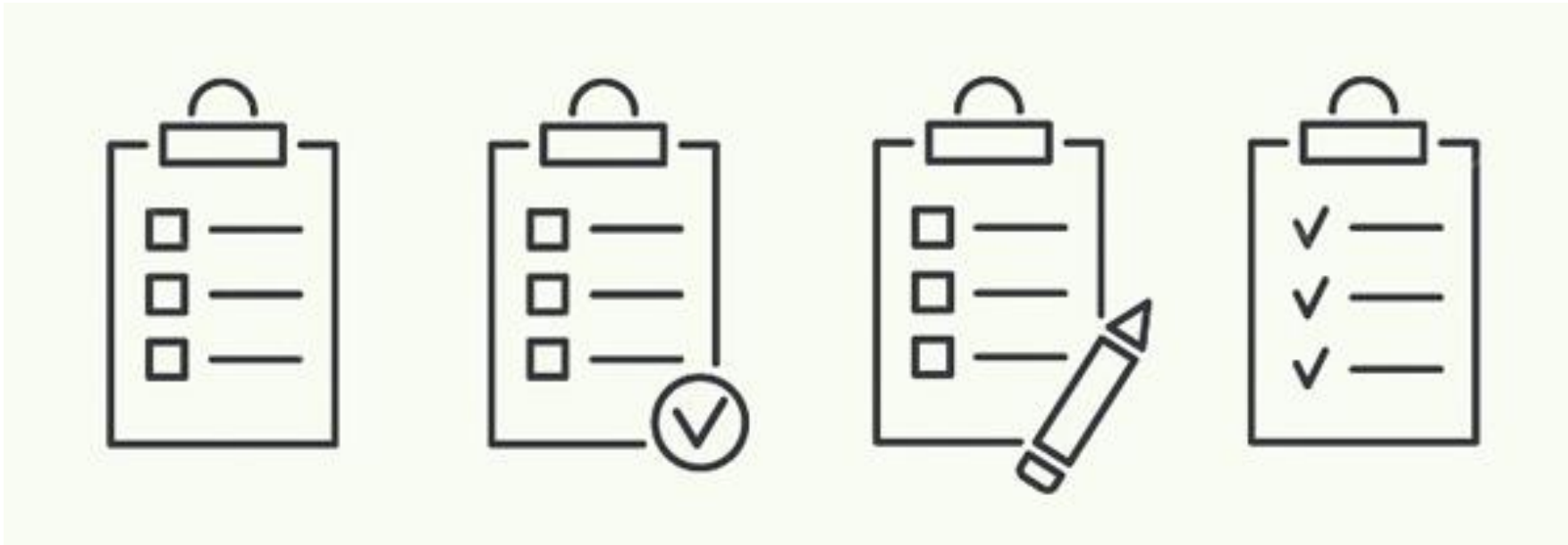


[[Math is Fun](#)]



Procedure

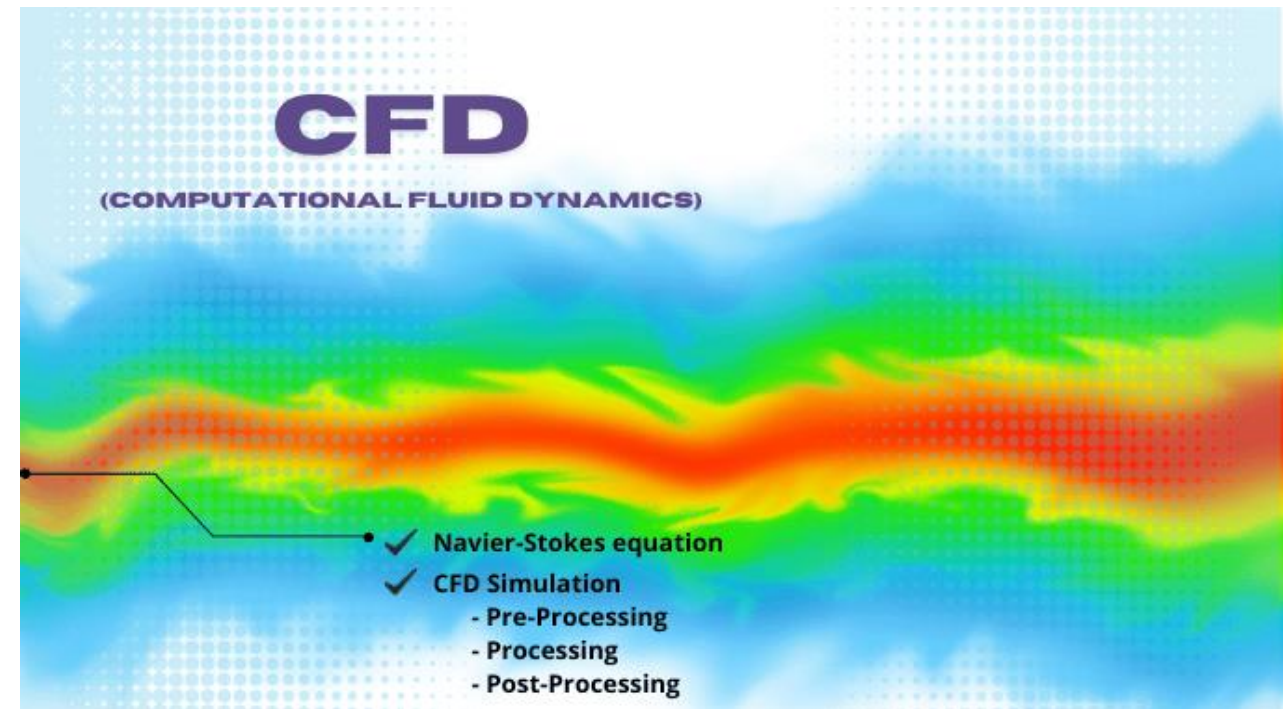
- Develop CFD simulations, model capsule options
- Experiment to validate
- Make conclusions and generalizations



CFD Modeling

Computational Fluid Dynamics

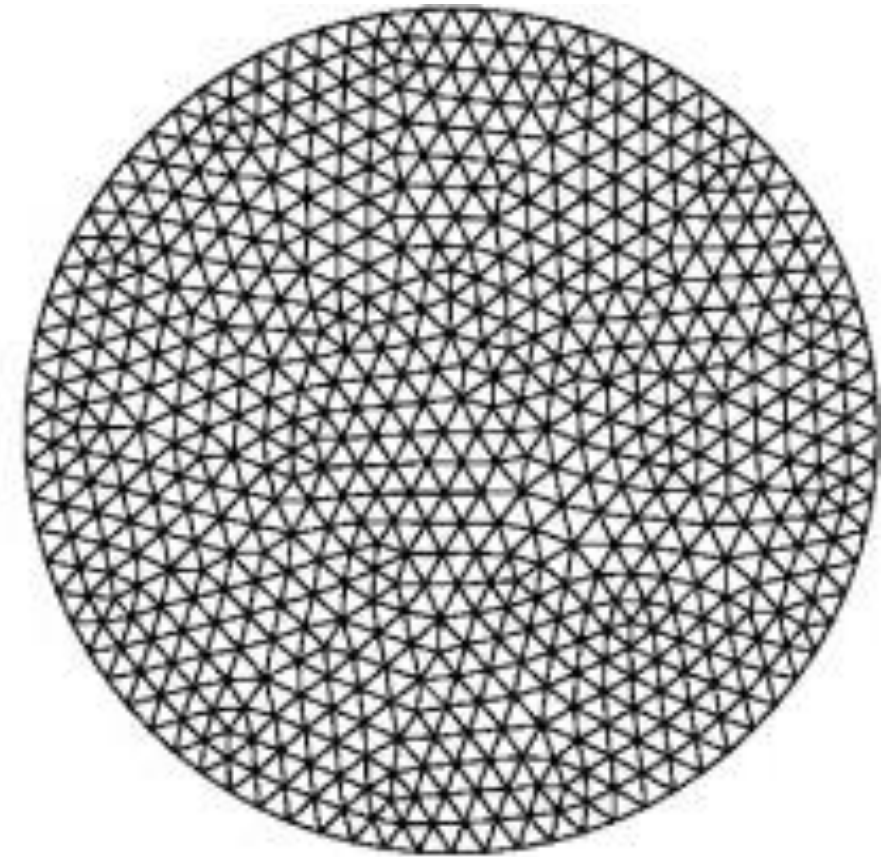
- **Computational Fluid Dynamics (CFD)**: analysis of fluid flow and heat transfer.
- Three fundamental equations governing fluid motion:
 - Conservation of **mass**
 - Conservation of **momentum** (Newton's 2nd law)
 - Conservation of **energy**



[[Mr CF](#)]

CFD Characteristics

- **CFD Analysis:** pre-processing, processing, post-processing
- **Solver:** finite difference, finite element, or spectral methods.
- Working results are characterized by **convergence** and grid independence.
- **Accuracy** and **cost** of any CFD solution is based on the number of **cells** chosen in the grid.



[[ScienceDirect](#)]

Enthalpy Porosity Method

Fundamental equations to simulate fluid flow and heat transfer:

CONSERVATION OF MASS $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) = 0$

MOMENTUM
CONSERVATION $\rho \frac{\partial \vec{u}}{\partial t} + \rho (\vec{u} \cdot \nabla) \vec{u} = \nabla \cdot [-p \mathbf{I} + \mathbf{K}] + \vec{S}$

CONSERVATION OF ENERGY $\rho C_p \frac{\partial T}{\partial t} + \rho C_p \vec{u} \cdot \nabla T = \nabla \cdot (k \nabla T)$

Mushy zone apparent heat capacity: $C_p = \frac{1}{\rho} (\theta_1 \rho_1 C_{p,1} + \theta_2 \rho_2 C_{p,2}) + L_{1 \rightarrow 2} \frac{\partial \alpha_m}{\partial T}$

Enthalpy Porosity Method

The **momentum source term** $\vec{S}$: $\vec{S} = \rho \vec{g} + A(\gamma) \vec{u}$

The porosity function:
[0 for liquid, 1 for solid]

$$A(\gamma) = \frac{C(1 - \gamma)^2}{\gamma^3 + \varepsilon}$$

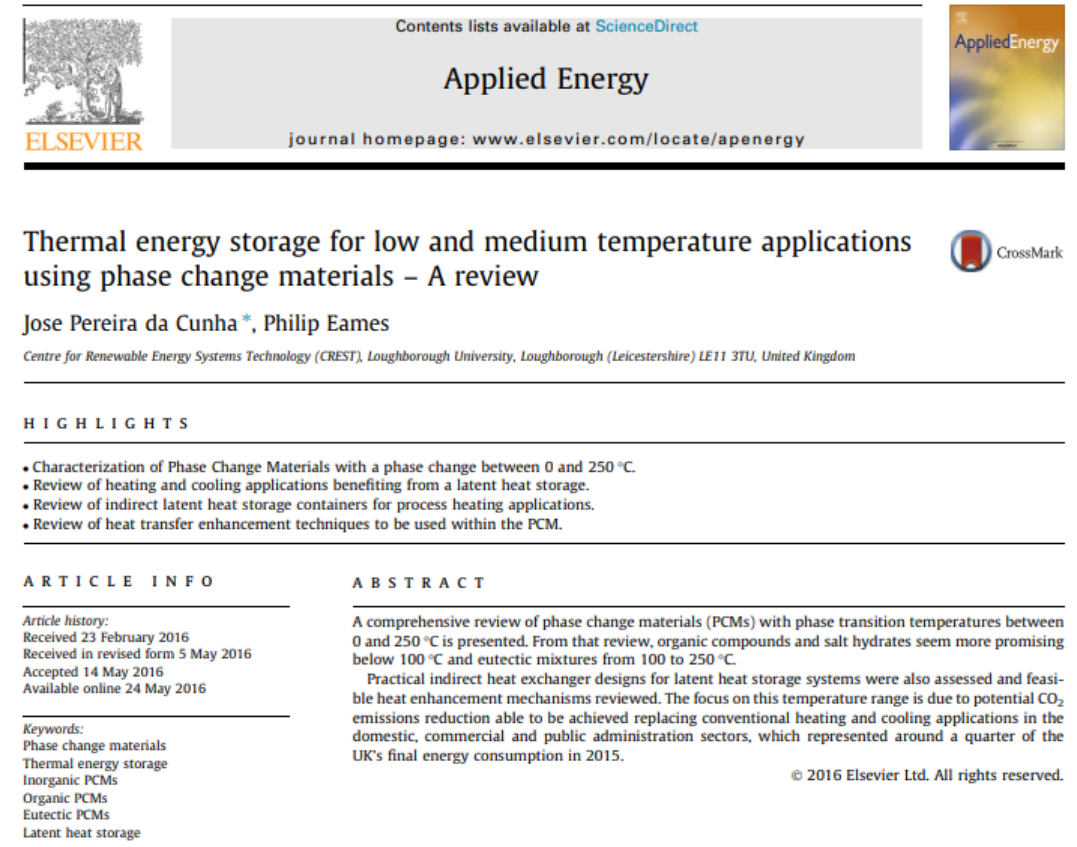
Melt fraction given by:
[0 for solid, 1 for liquid]

$$\gamma = \begin{cases} 0 & \text{if } T < T_{sol} \\ \frac{T - T_{sol}}{T_{liq} - T_{sol}} & \text{if } T_{sol} \leq T \leq T_{liq} \\ 1 & \text{if } T > T_{liq} \end{cases}$$

$$\gamma = \alpha + 0.5$$

CFD Simulation Literature

- The agreement between numerical and experimental results has been well-established in literature:
 - 2017 Bellan, 2015 Bellan, 2014 Archibold (spherical)
 - 2022 Yan, 2021 Mallya (cylindrical)
 - & others



[[Pereira da Cunha, 2016](#)]

COMSOL Software

- **COMSOL**: finite element method, user-friendly, hard to debug
- Analyze, model, and animate.

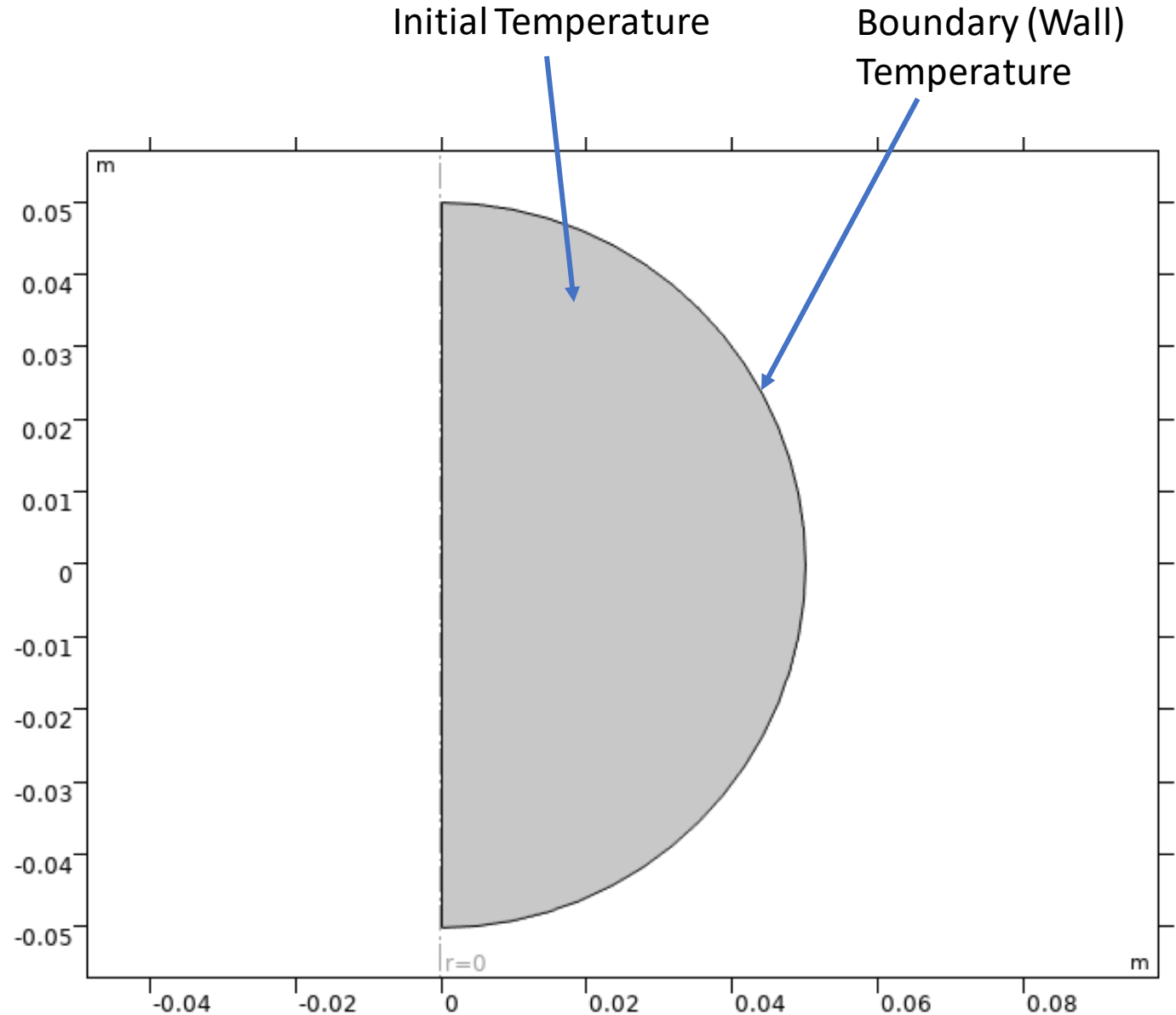


[[Jamali, 2021](#)]

COMSOL PCM Simulations

Model

- **2D Axisymmetric:** **semicircle**, using a time-dependent laminar flow study.
- Assumptions and simplifications



COMSOL Simulation Path

The COMSOL gradual **simulation path** included:

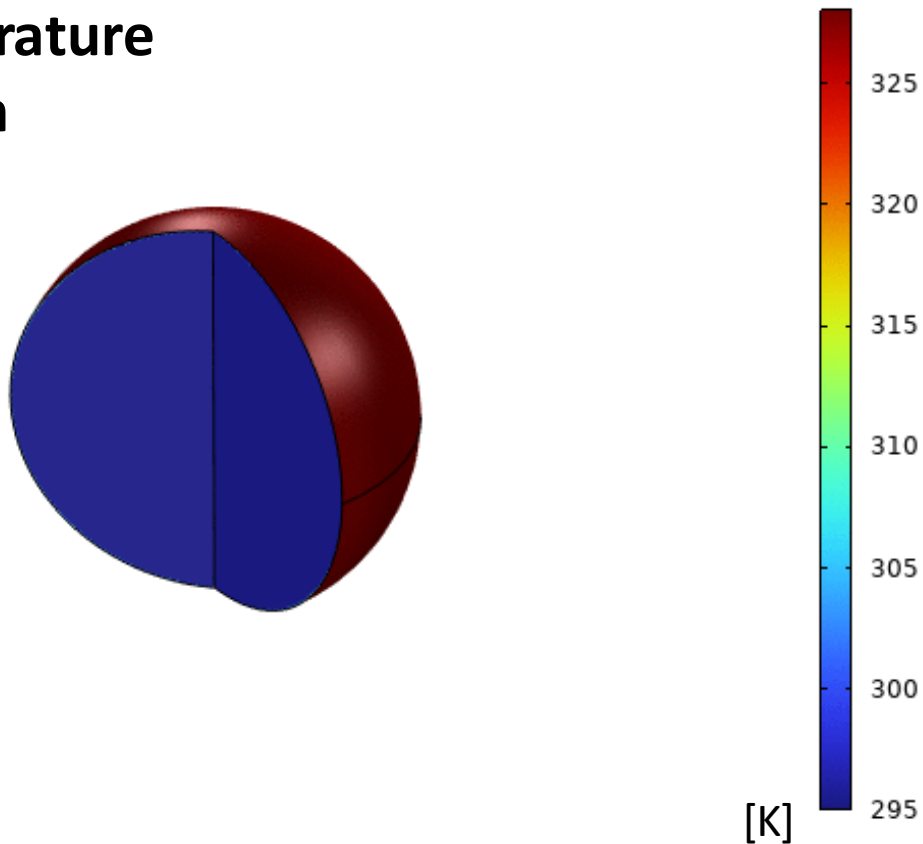
1. Sphere, filled with water
2. Individual exploration
3. Phase change physics and new materials
4. Molten Salt Spherical PCM Capsule MELTING
5. Molten Salt Spherical PCM Capsule SOLIDIFICATION
6. Potential Experiment Simulations (Urea, Paraffin Wax, Water)

Our **end goal** was for the simulation to accurately represent the PCM capsule.

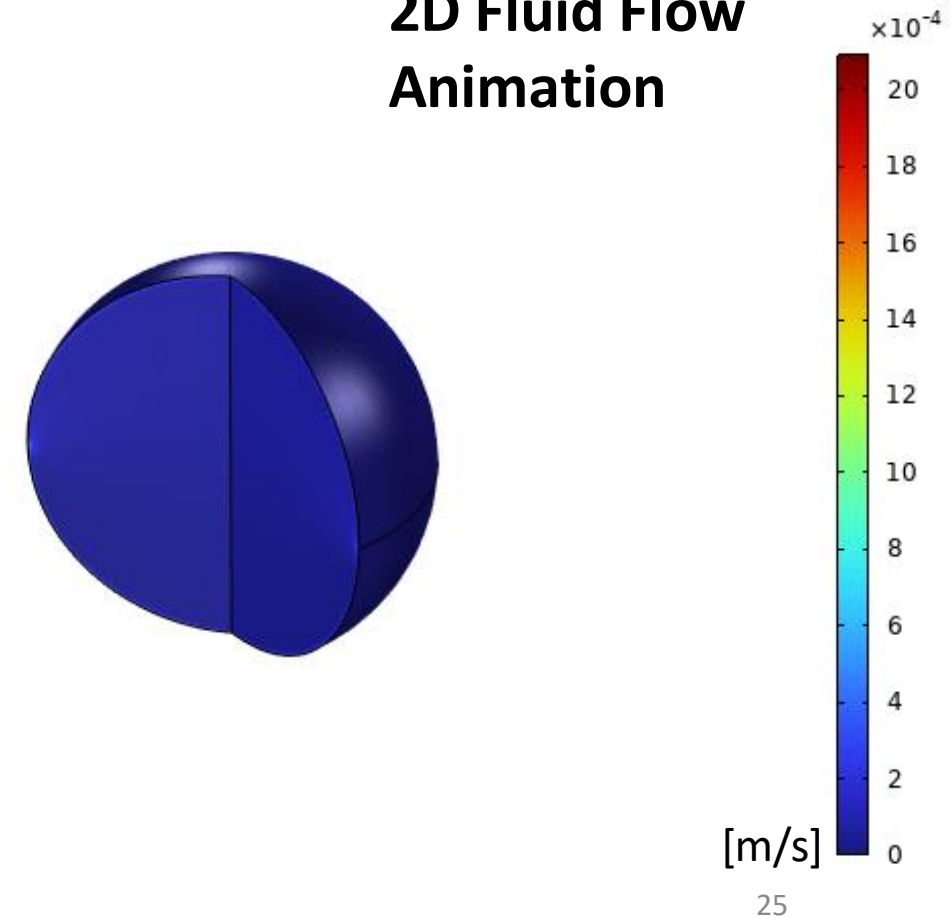
1. Sphere with Water

360 seconds

**3D Temperature
Animation**



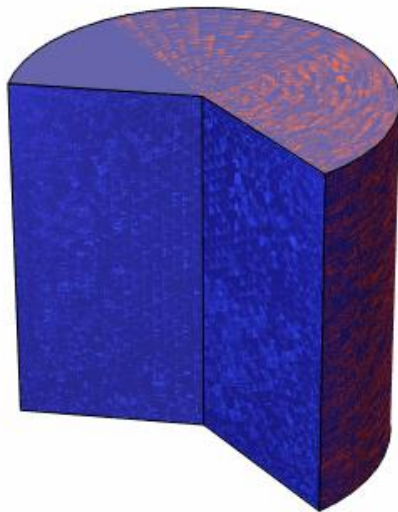
**2D Fluid Flow
Animation**



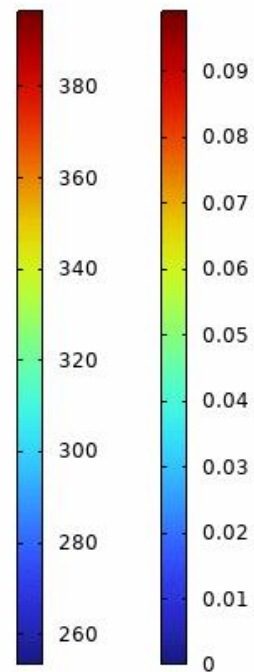
2. Individual Exploration

Mercury Cylinder:

20 seconds

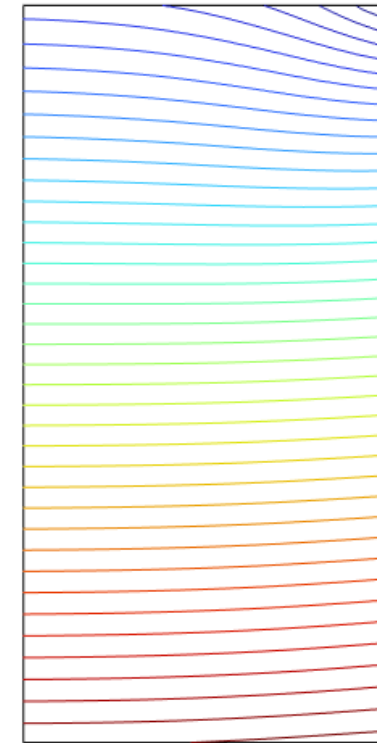


**3D Fluid Flow + Temp
Surface Animation**

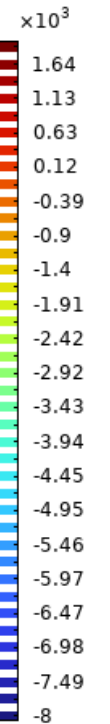


[K] [m/s]

2D Pressure Screenshot



[atm] (reference pressure: 1 atm)

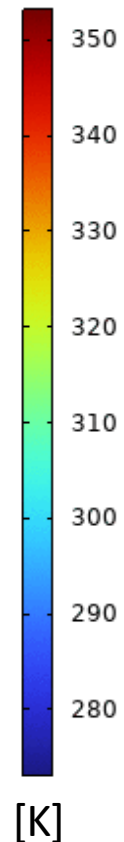
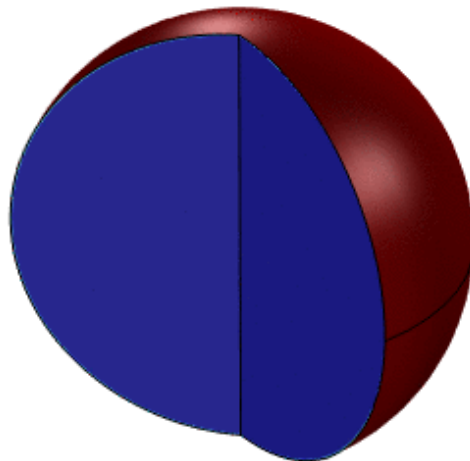


2. Individual Exploration

Sphere with gasoline:

**3D Temperature
Animation**

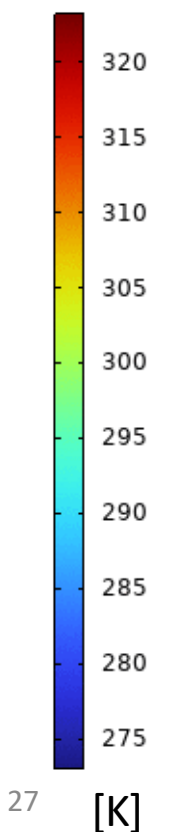
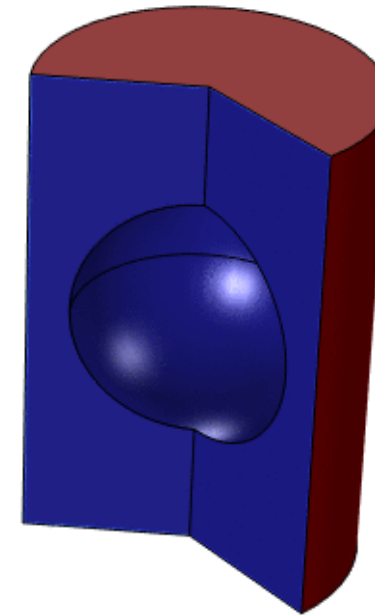
1800 seconds



Hollow cylinder with hydrogen:

**3D Temperature
Animation**

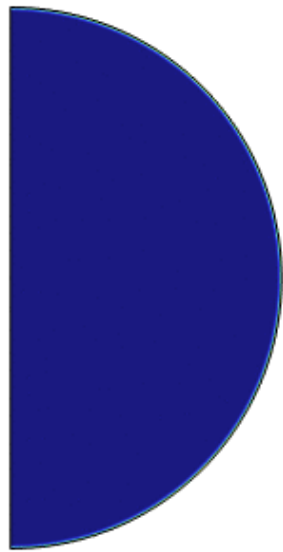
1 second



3.1 Phase Change Physics

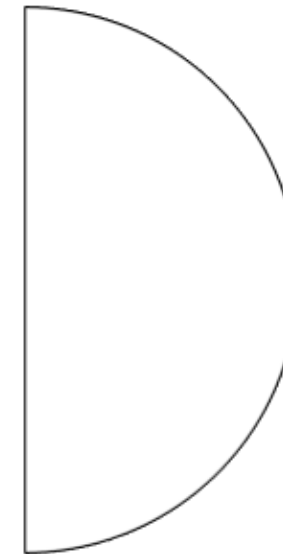
Phase Change (ice and water):

**2D Temperature
Animation**



2000 seconds

**2D Melt
Fraction
Animation**

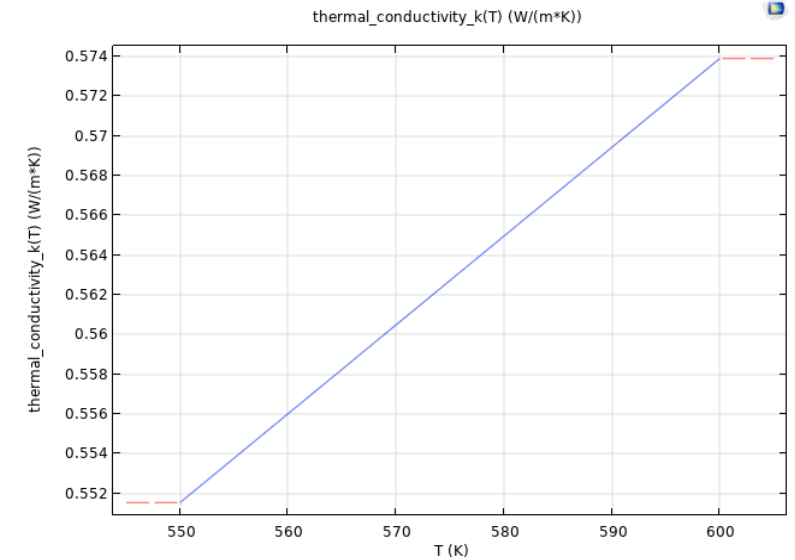


3.2 New Material

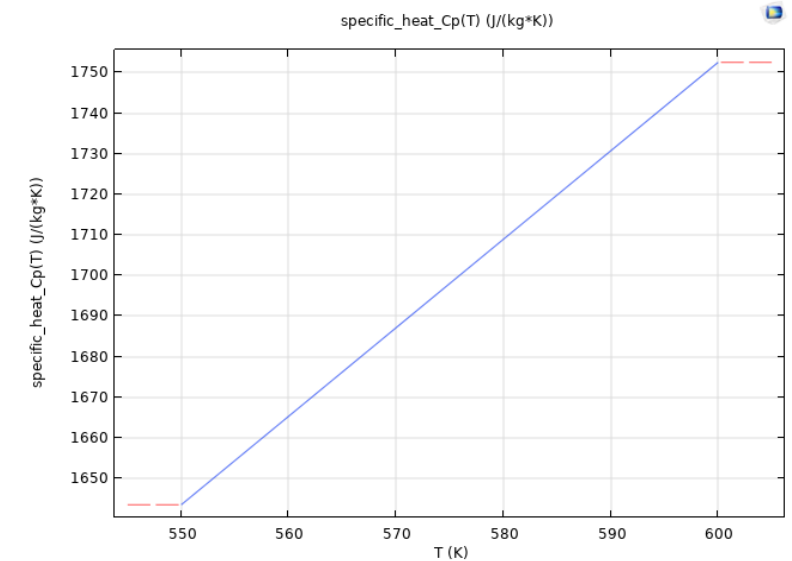
New material (Sodium Nitrate):

- New equations
- Density linear mushy zone interval

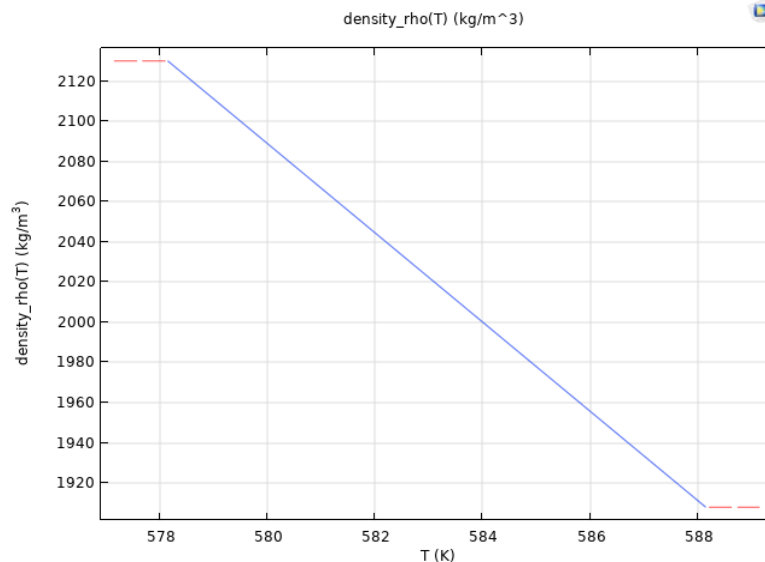
Thermal Conductivity



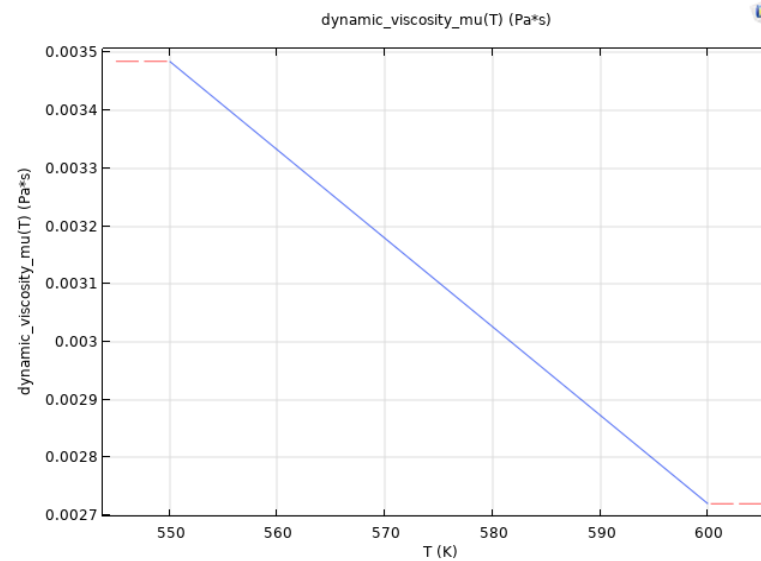
Specific Heat



Density

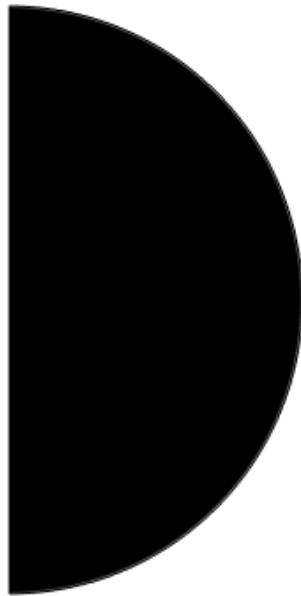


Dynamic Viscosity

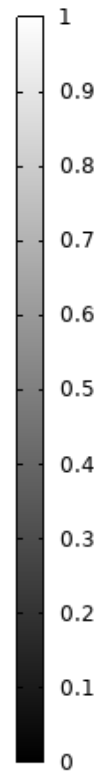


4. Molten Salt Spherical PCM Capsule Melting

**2D Melt
Fraction
Animation**

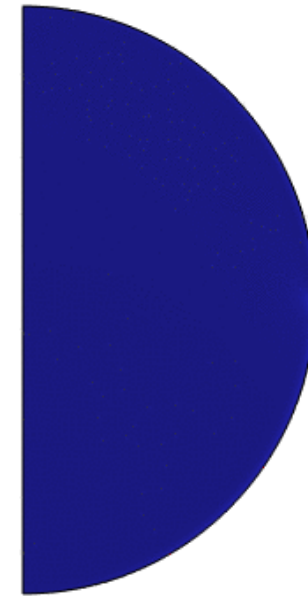


350 seconds



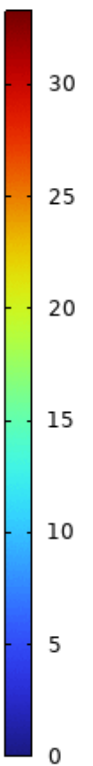
[-]

**2D Fluid Flow
Animation**



[m/s]

$\times 10^{-3}$



4. Version 2

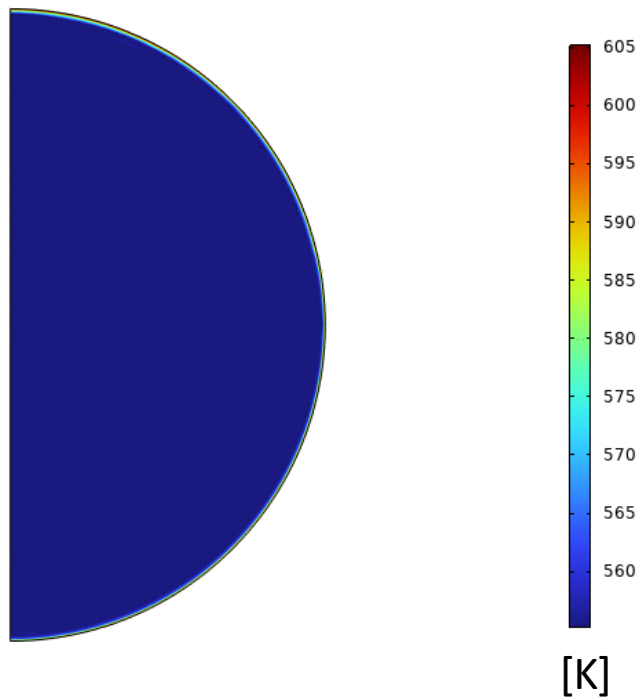
1800 seconds

Volume force:

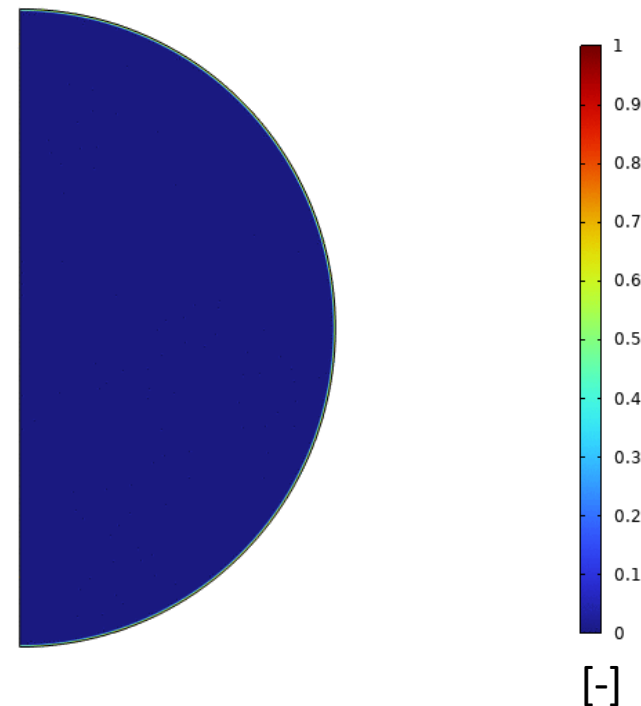
F	$-C \cdot (1 - (\text{ht.alpham12} + 0.5))^2 / ((\text{ht.alpham12} + 0.5)^3 + \text{epsilon}) \cdot u$	r	N/m ³
	$-C \cdot (1 - (\text{ht.alpham12} + 0.5))^2 / ((\text{ht.alpham12} + 0.5)^3 + \text{epsilon}) \cdot w$	z	

Name	Expression	Value	Description
C	108[kg/m ³ /s]	108 kg/(m ³ ·s)	melting front morphology constant
epsilon	.001	0.001	computational constant

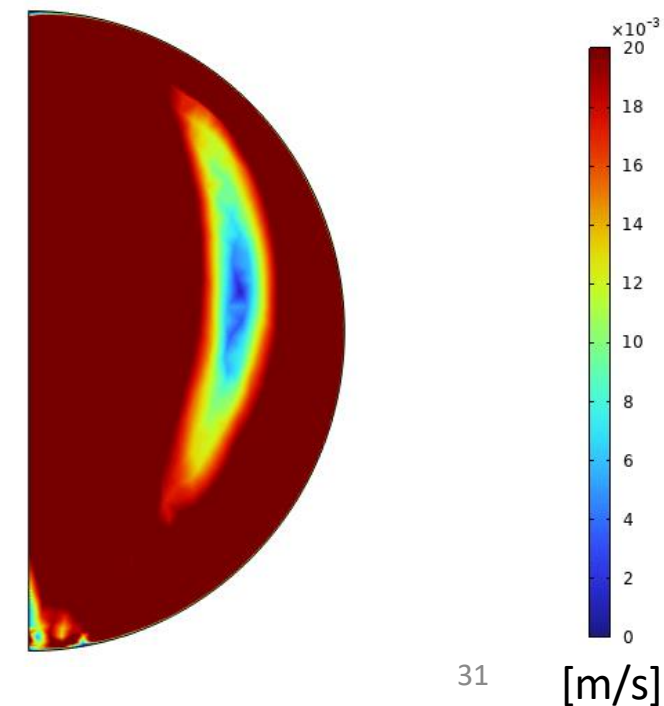
2D Temperature



2D Melt Fraction

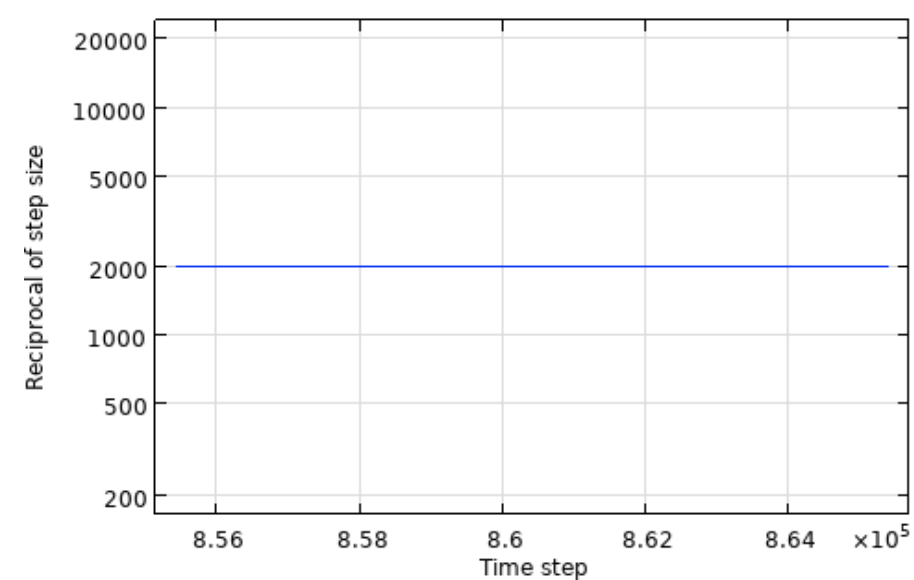
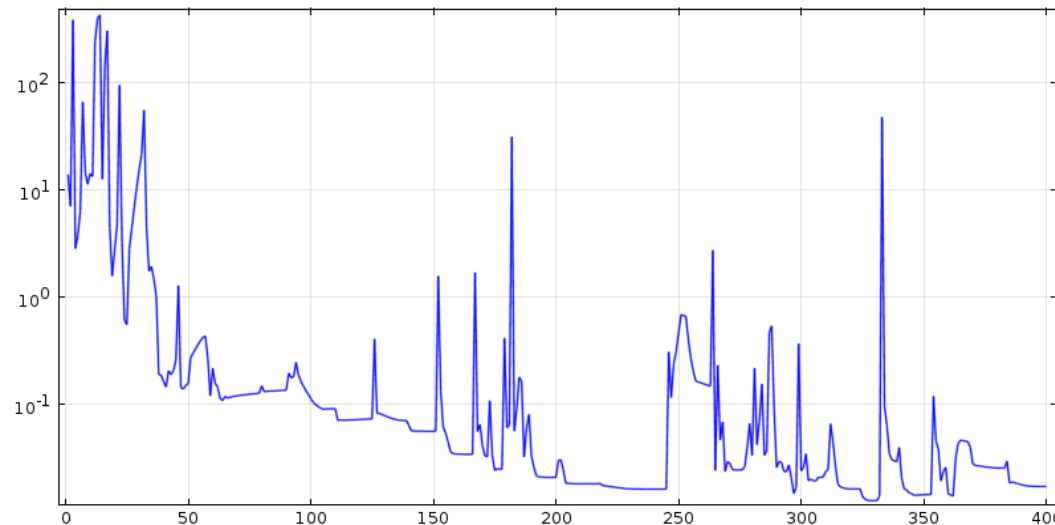


2D Fluid Flow

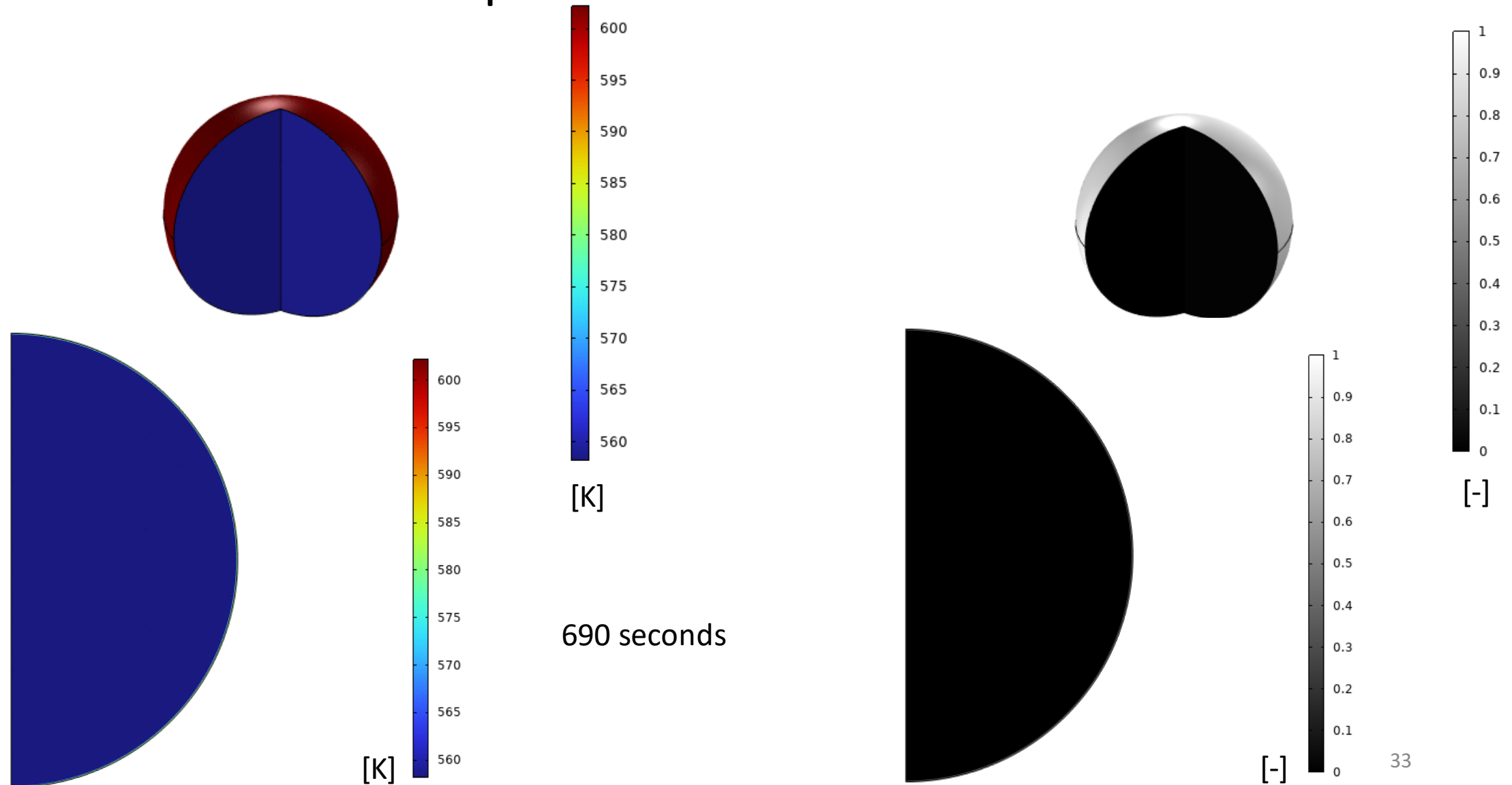


4. Version 3, 4, 5 (Optimizations)

- The **Dynamic Viscosity** (μ) property plot was then modified to account for different phases of the material.
- Further **optimizations** were integrated, including, but not limited to, the timesteps, the tolerance factor, the maximum number of iterations, the mesh, and the mushy zone temperature interval.



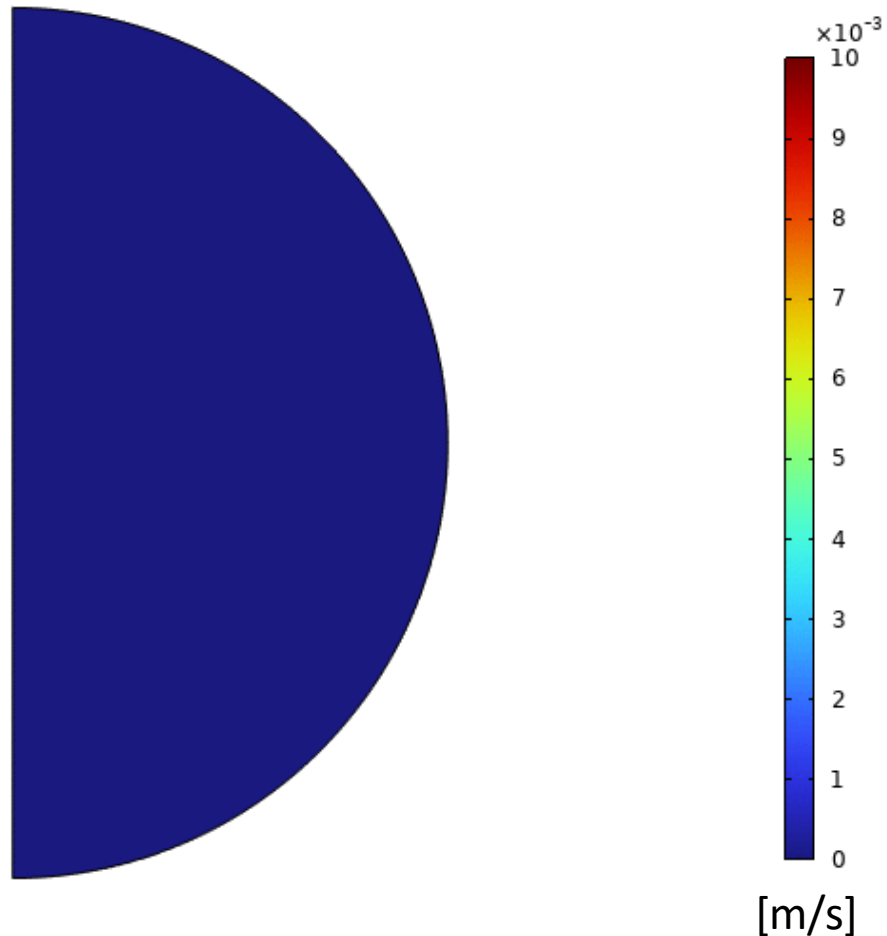
4. Version 5: Temperature & Melt Fraction



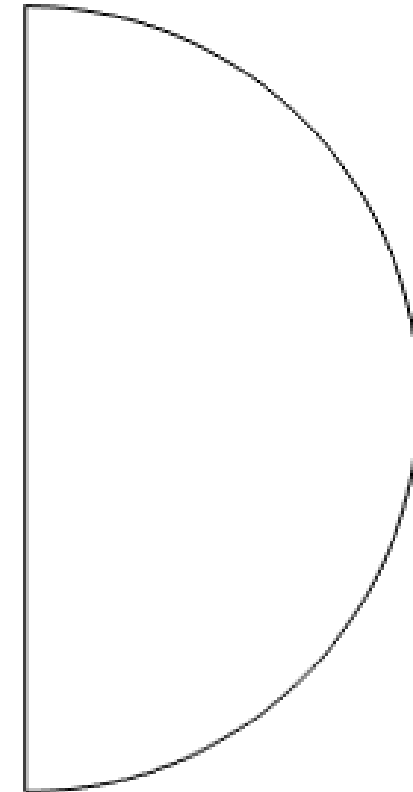
4. Version 5 Results: 2D Fluid Flow

2D Fluid Flow, Rainbow Surface

690 seconds



2D Fluid Flow, Magenta Arrow Surface (SCALED x2)



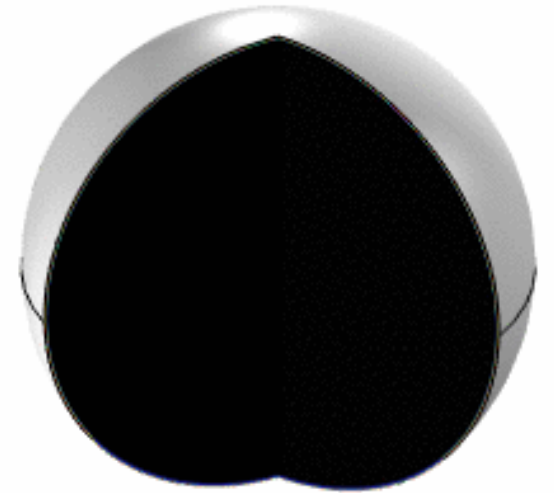
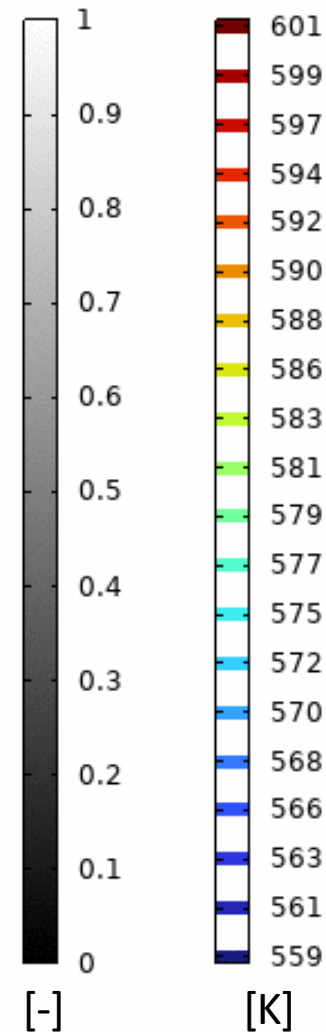
4. Version 5 Results: Combined

690 seconds

**Greyscale Surface:
Melt Fraction [-]**

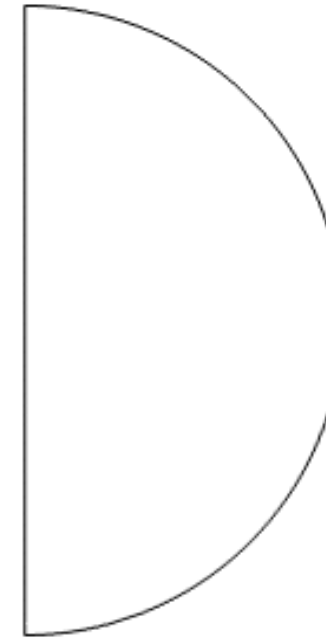
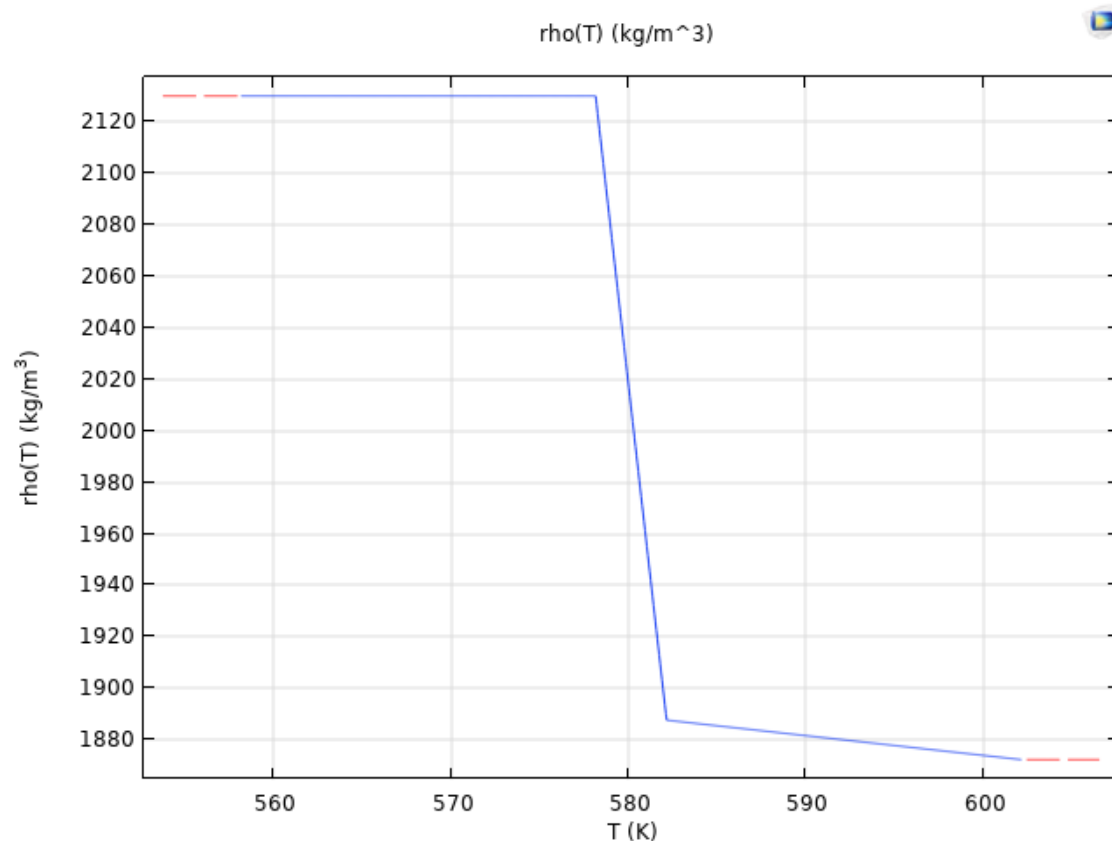
**Rainbow
(Isothermal) Contour:
Temperature [K]**

**Magenta Arrows:
Fluid Flow Velocity
(SCALED x2) [m/s]**

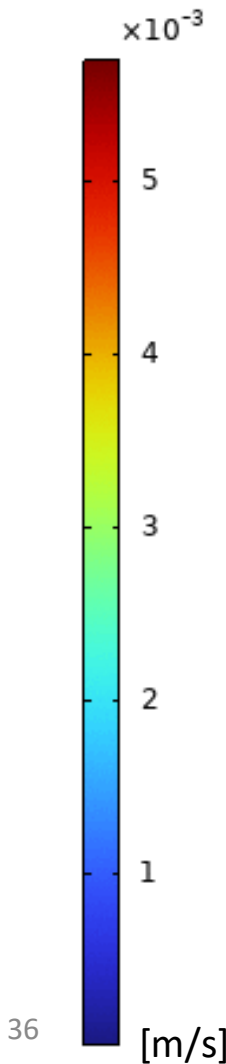


4. Version 6 Results: Recirculation

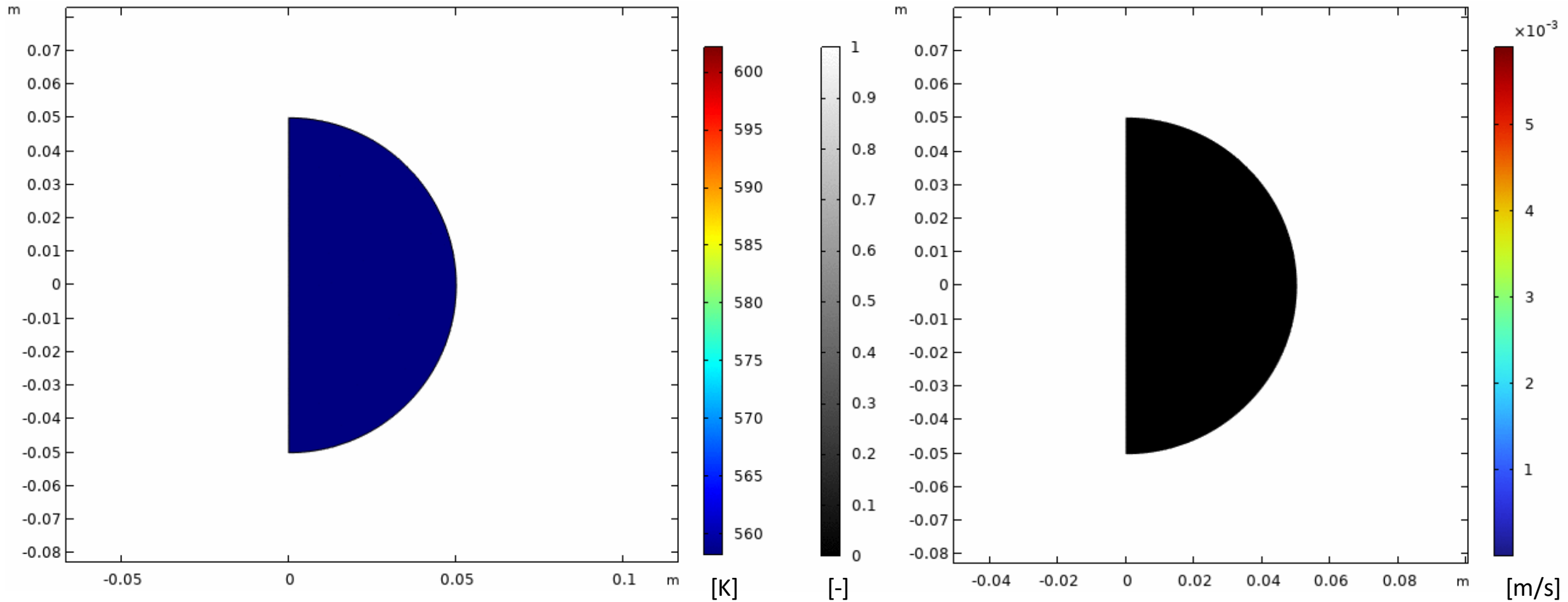
- Temperature dependent liquid density, C value: 10^4
- Recirculation trends



980 seconds



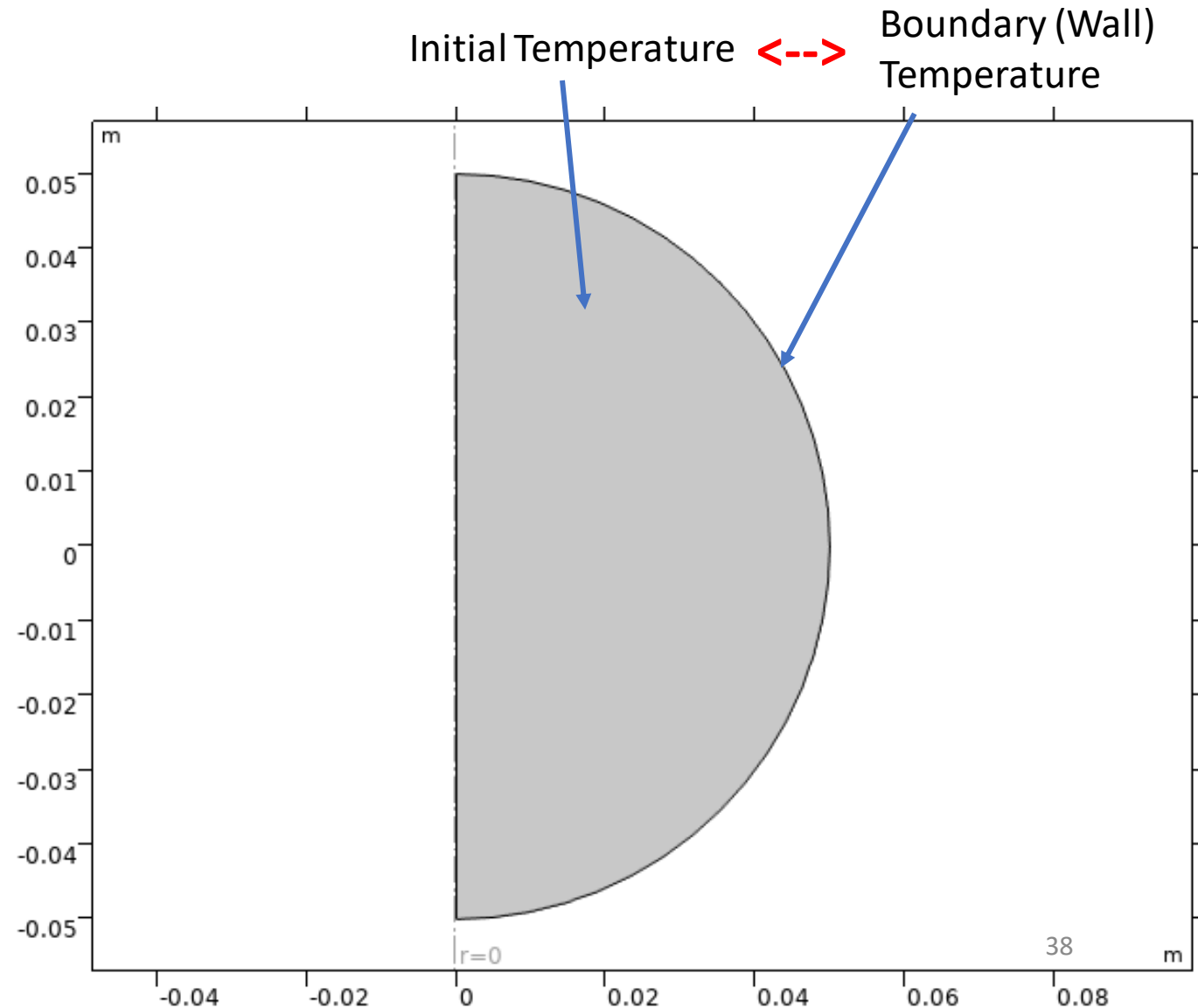
4. Version 6 Results: Recirculation



980 seconds

5. Molten Salt Spherical PCM Capsule Solidification

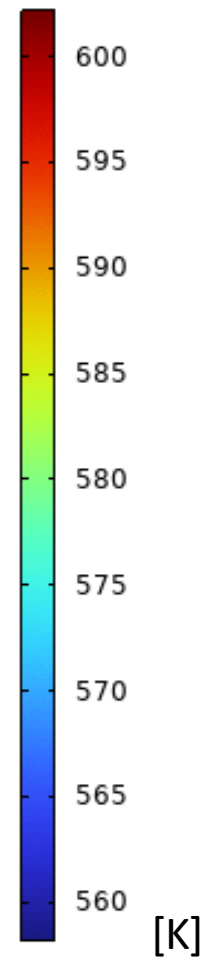
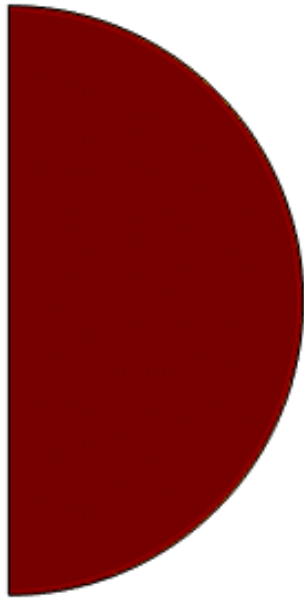
- To model the transformation from **liquid to solid**, we switch the initial and boundary temperatures.



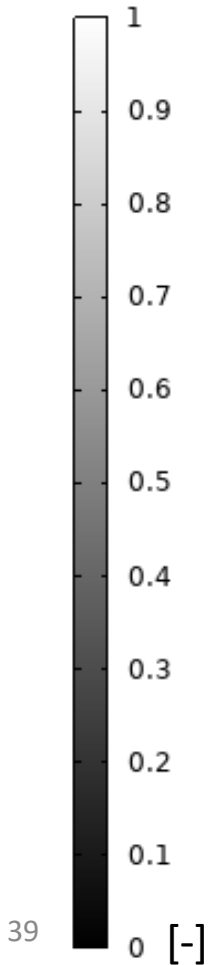
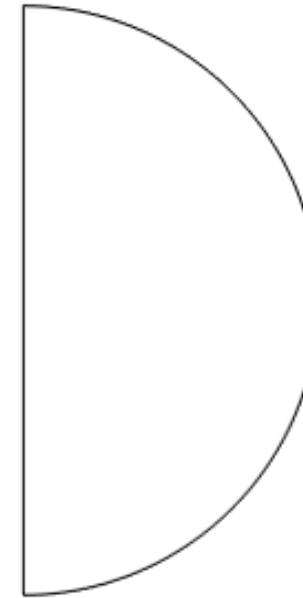
5. Solidification Version

1800 seconds

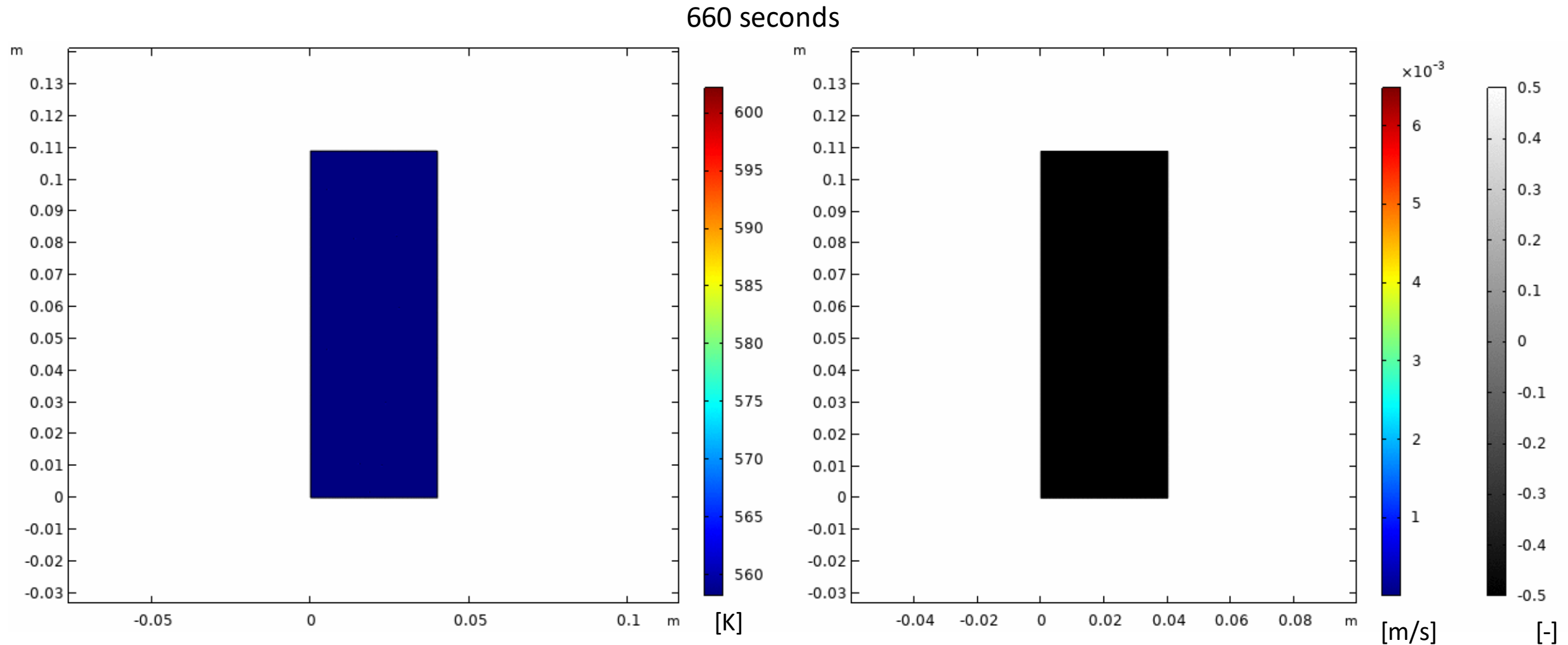
**2D Temperature
Animation**



**2D Melt Fraction
Animation**

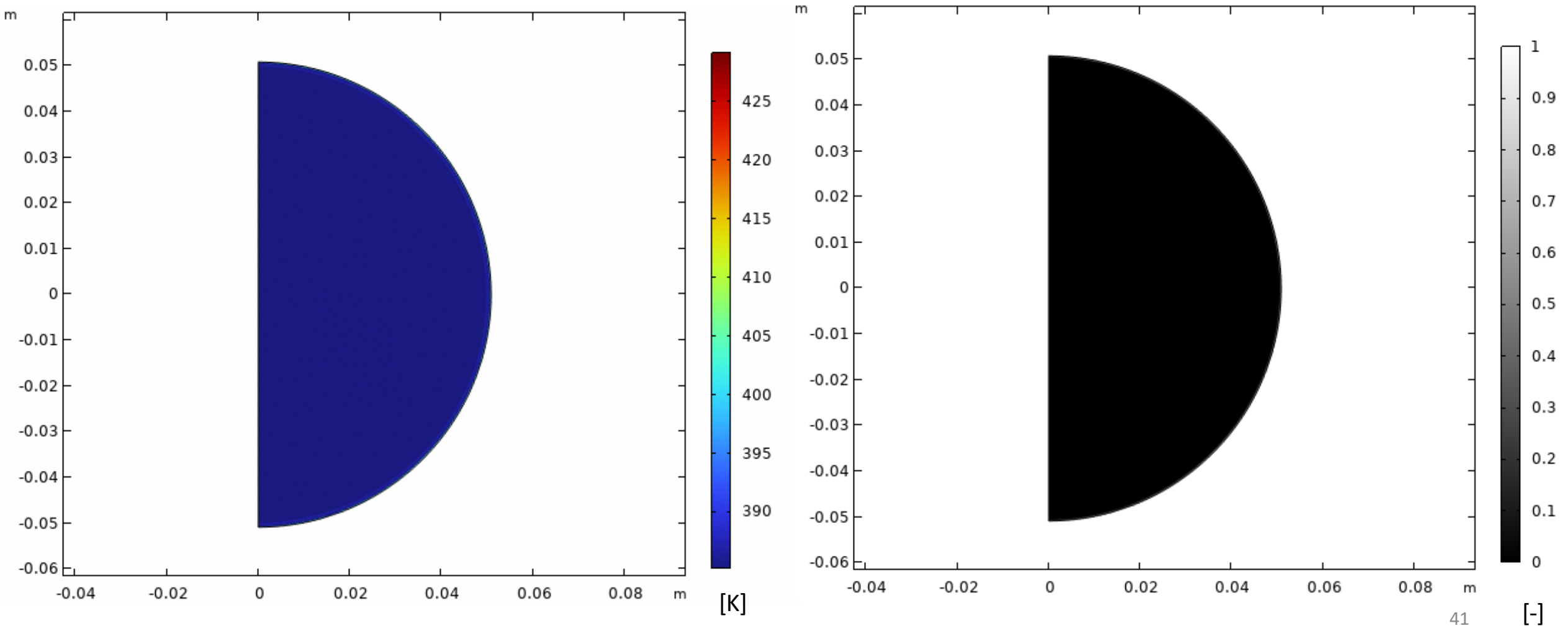


6. Molten Salt Melting: Different Geometries



6. Urea: Spherical Capsule

240 seconds



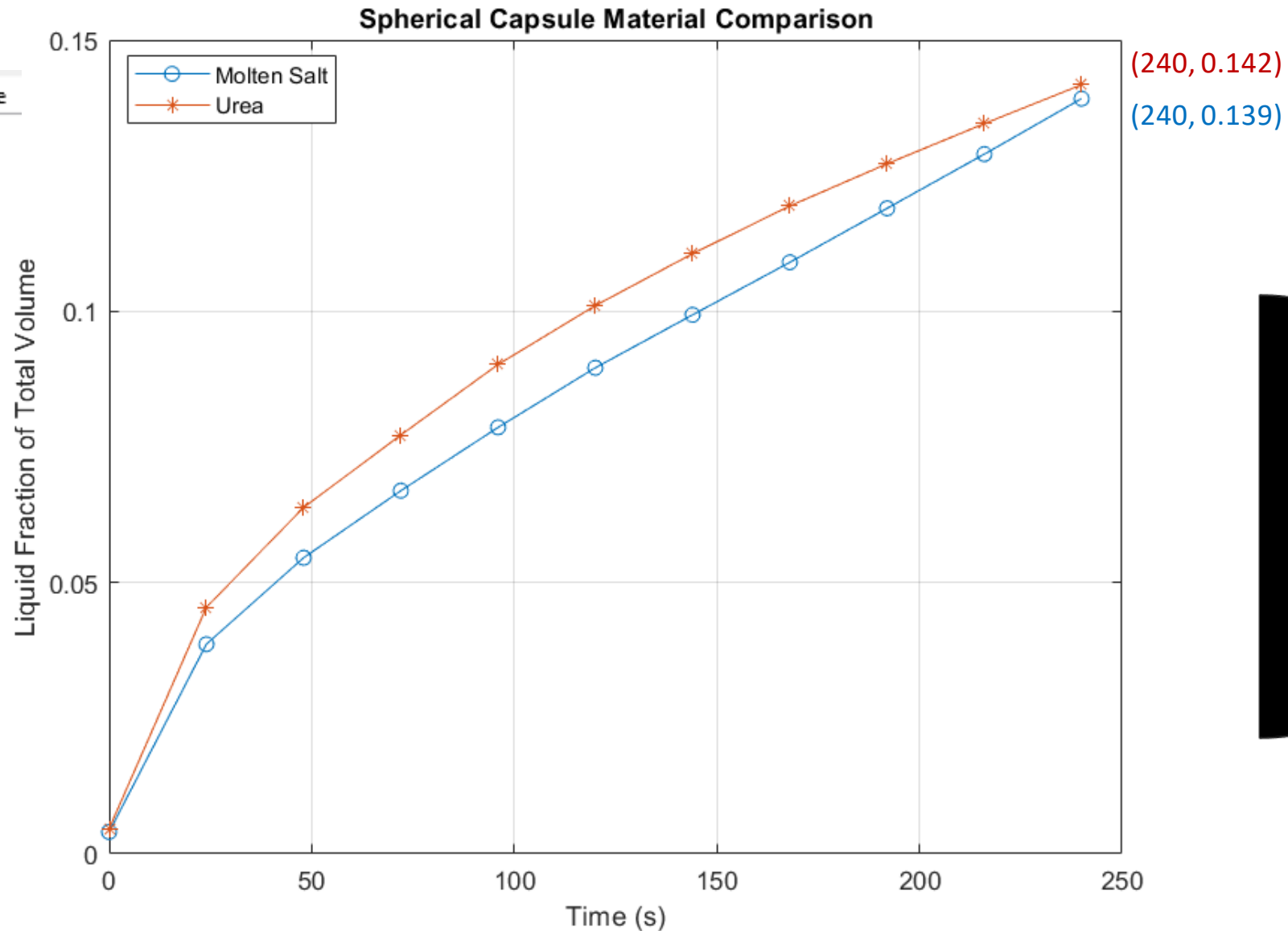
Material Comparisons

240 seconds

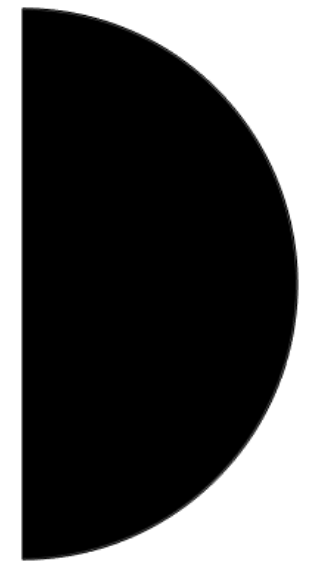
Volume integration:

```
if(ht.alpham12>0, 1, 0)/capsule_volume
```

Molten Salt

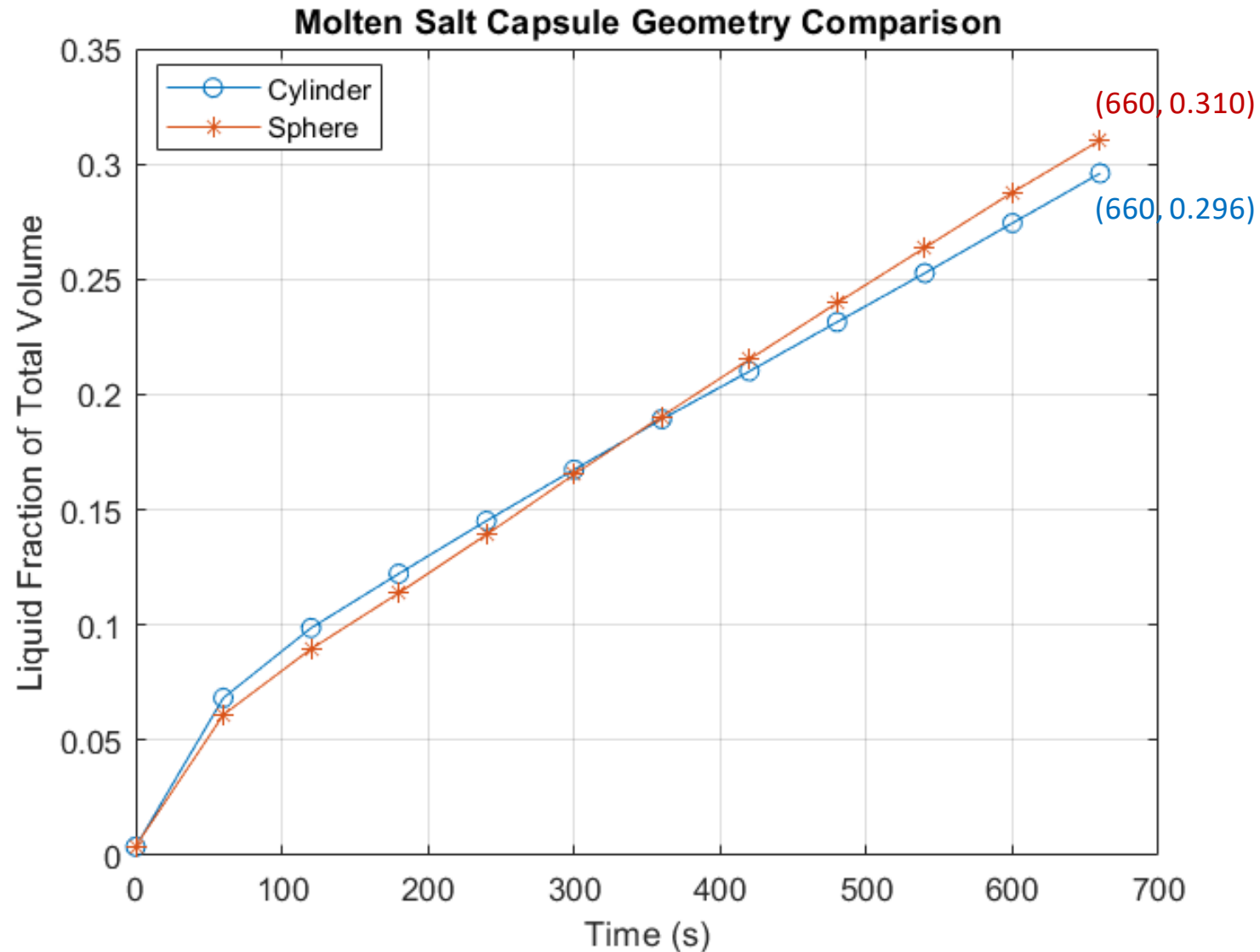


Urea



Geometry Comparisons: Molten Salt

660 seconds



Potential Concerns and Uncertainties

- Melting front morphology constant value
- Assumption that melting temperature = solidification temperature (phase change hysteresis)
- Number and thickness of boundary layers used
- Timestep used, congruency with mesh size
- Pressure point allocations
- Viscosity value for the solid phase
- Mushy zone temperature interval (4[K] instead of 2[K])

CFD Model Validation Experiment

Experiment Plan & Method

Plan:

- Use **Paraffin Wax** (PW) to represent the Urea PCM capsule, and scale accordingly.
- Validate CFD results for other models using the experiment.



[[ThoughtCo](#)]



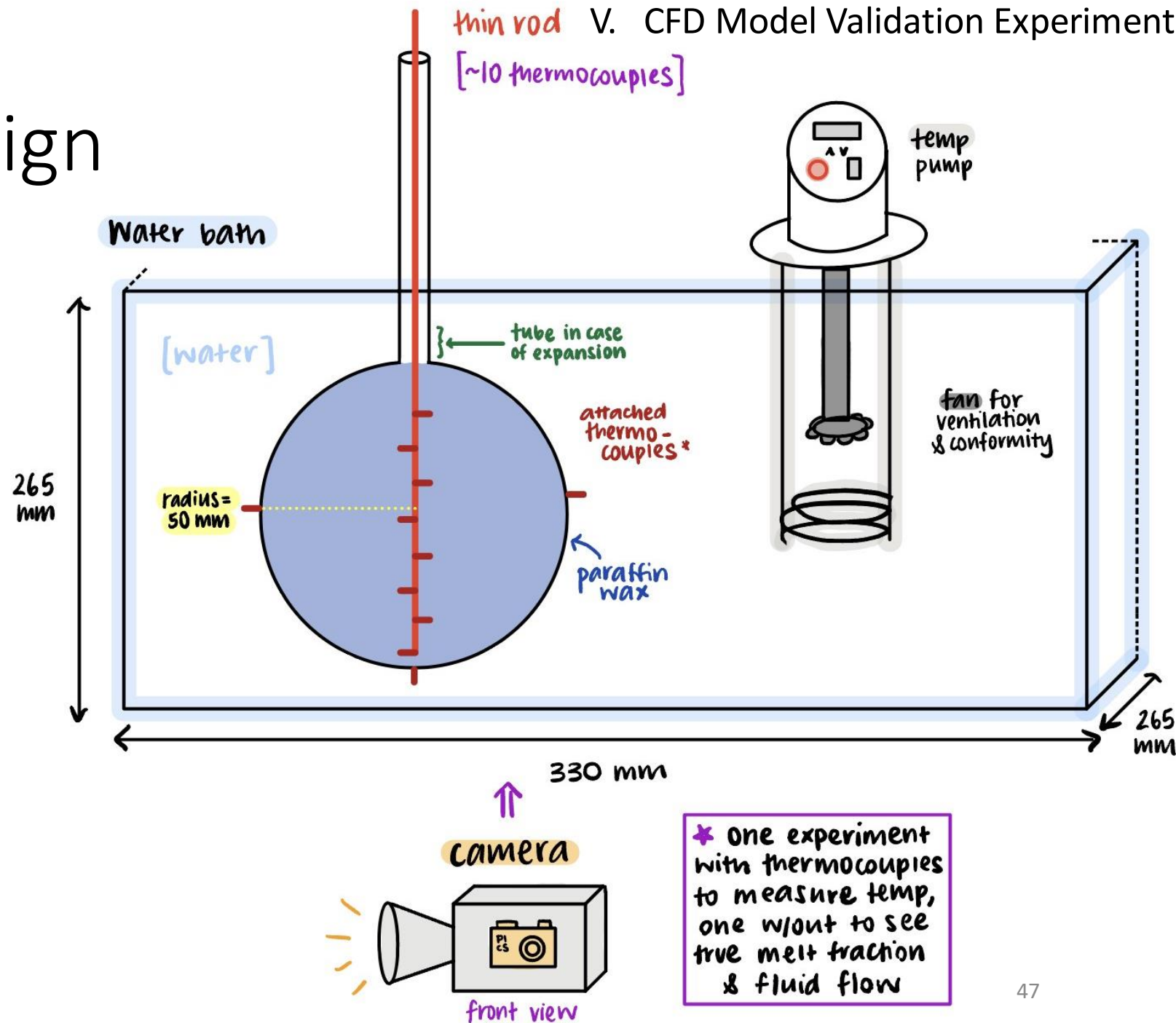
[[5.12 Solutions](#)]

Experiment Design

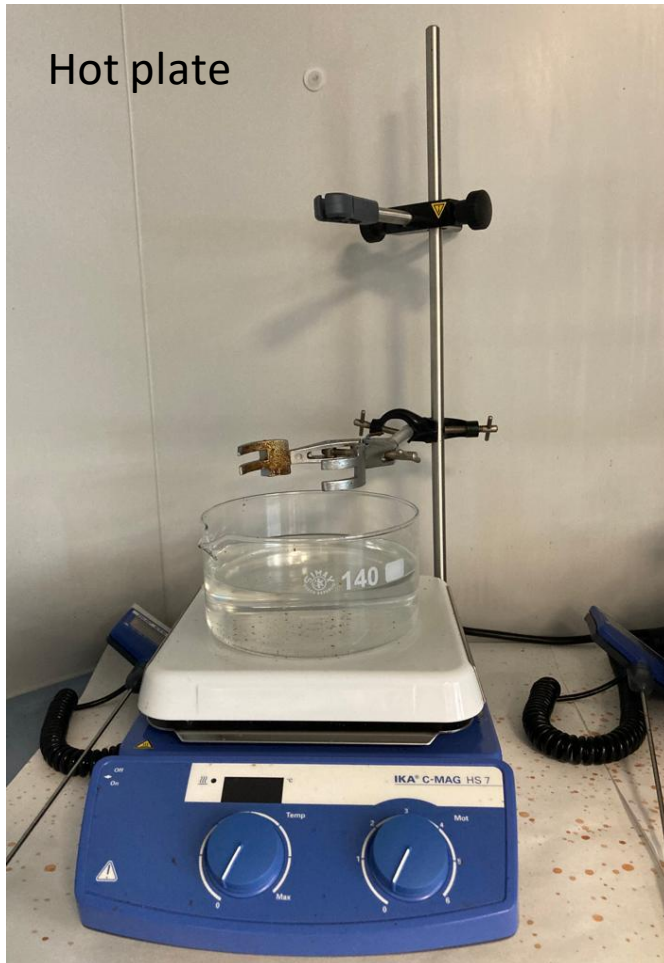
Measurement:

- **Thermocouples** to measure temperature
- **Camera** to measure the melt fraction

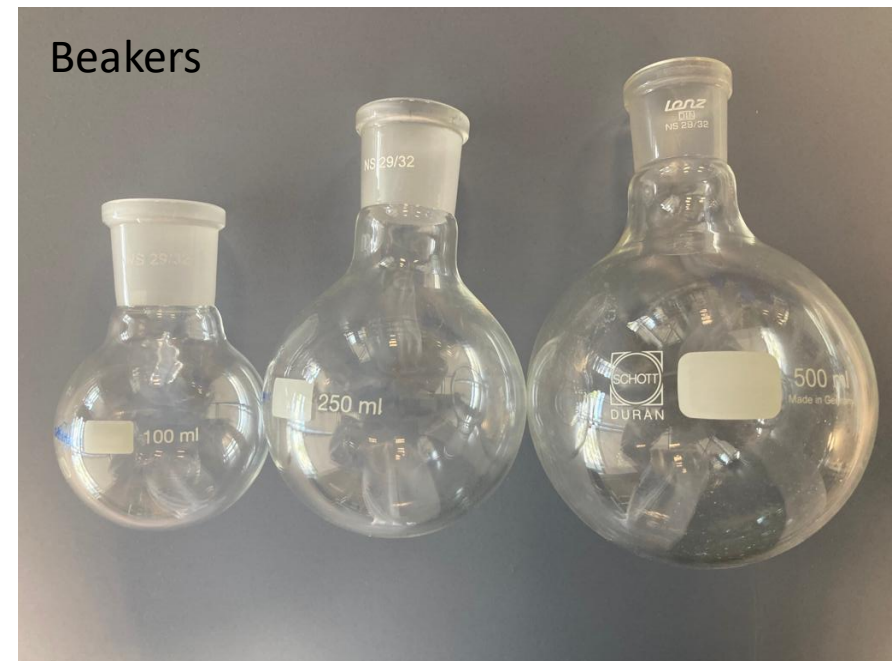
Compare the numerical and experimental results and apply to Urea model.



Lab Materials

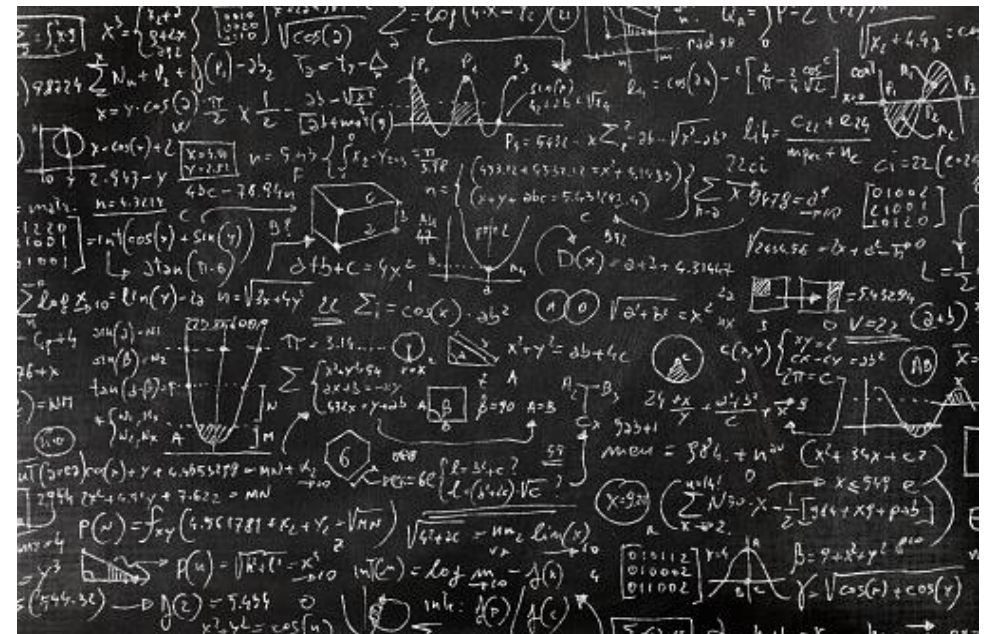


Infrared camera



Dimensionless Numbers

- Initial priority: similar dimensionless numbers between the experiment and final material
- Updated: wax simulation compared to wax experimental results



Paraffin Wax & Water Simulations

- Regular model and constrained model for the paraffin wax melting simulation
- Constrained model for the water/ice phase change simulation



[[Fundamental Photographs, 2015](http://www.fphoto.com)]

Next Steps

- Experiment simulations
 - Constrained and unconstrained paraffin wax experiment model
 - Constrained water/ice experiment model
- Conduct the experiment(s)
- Validate results
- Remaining simulations:
 - Effect of square geometry on all materials
 - Effect of cylindrical geometry on wax and water
 - Refined mesh for all critical results
 - Incorporate an air gap and a solid material boundary

Conclusions

- Effect of materials and effect of geometries
- My takeaways
- Applications



Madrid Personal Experience

Madrid Experience

- Unique experience, independent summer.
- Weekend travel.
- The architecture, food, atmosphere, and heat.
- Beach volleyball!
- Spanish language.
- Hoping to come back soon!



iGracias!
Thank you all!

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Questions?